Online Hospital Management System for GenNet

Sabikun Nahar Lima

ID# 22103191

A Practicum in the Partial Fulfillment of the Requirements

for the Award of Bachelor of Computer Science and Engineering (BCSE)

Department of Computer Science and Engineering

College of Engineering and Technology

IUBAT—International University of Business Agriculture and Technology

Fall 2025

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The practicum has been examined and approved,

Prof. Dr. Utpal Kanti Das
Chairman

Shahinur Alam
Co-supervisor, Coordinator and Assistant Professor

Jannatul Ferdous
Supervisor and Lecturer
Department of Computer Science and Engineering
College of Engineering and Technology
IUBAT—International University of Business Agriculture and Technology

Fall 2025

Letter of Transmittal

17 January 2026

The Chair

Practicum Defense Committee

Department of Computer Science and Engineering

IUBAT—International University of Business Agriculture and Technology

4 Embankment Drive Road, Sector 10, Uttara Model Town

Dhaka 1230, Bangladesh.

Subject: Letter of Transmittal.

Dear Sir,

With due respect, I am glad to submit my practicum report allow "Online Hospital Management System", developed as a partial requirement for the Bachelor of Computer Science and Engineering (BCSE) degree at IUBAT. During my three-month practicum at GENNET, I was involved in the evaluation, design, and development of a online hospital management system aimed at improving efficiency in patient registration, appointment scheduling, billing, and medical record management. The work accomplished during this practicum is reflected in this report.

I would like to express my sincere gratitude to the GENNET team for their cooperation and support throughout the practicum period.

Yours sincerely,

____

Sabikun Nahar Lima

22103191

Organization's Certificate

GenNet

Dated: 15th January, 2026

TO WHOM IT MAY CONCERN

This is to certify that **Sabikun Nahar Lima**, has successfully completed her internship in our organization from **12th October 2025 to 4th January 2026** as Intern.

During her service tenure as a team member with GenNet, Management found her as honest, sincere, and hardworking with professional attitude.

We wish her a successful career in her future professional life.

NETWORK FOR THE GENERATION GenNet

Md. Subbir Chowdhury Topu
Assistant Manager, HR & Operation
GenNet (Sister Concern of Rahimafrooz Bangladesh Ltd.)

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Student's Declaration

I hereby certify that I have completed the practicum report titled " Online Hospital Management System" in partial satisfaction of the requirements for the Bachelor of Computer Science and Engineering (BCSE) degree at IUBAT. I additionally certify that this report is entirely original with no prior submissions for a degree, diploma, or certificate to any other school or organization. Each source of information used in this report is properly cited in the text and listed in the reference section. The conclusions and analysis in this report are based on my personal research and practical experiences from the practicum. Any mistakes or omissions in this essay are entirely my responsibility.

Sabikun Nahar Lima

22103191

Supervisor’s Certification

This is to certify that the practicum report titled “Online Hospital Management System” has been completed by Sabikun Nahar Lima, ID:22103191, at the International University of Business Agriculture and Technology (IUBAT) as a part of partial fulfillment of the requirements for the award of Bachelor of Computer Science and Engineering (BCSE) program. This report has been prepared under my supervision. The performance of Sabikun Nahar Lima is truly appreciable in terms of attendance, communication, problem-solving abilities and responsibility. Her project work, and final report have been revised, and confirmed that the practicum requirements have been satisfactorily fulfilled. She is now eligible to appear for the practicum defense. I wish her success and excellence for her future endeavors.

Jannatul Ferdous
Supervisor and Lecturer
Department of Computer Science and Engineering
IUBAT—International University of Business Agriculture and Technology

Abstract

Hospital management systems play a vital role in streamlining hospital operations, maintaining accurate medical records and ensuring timely access to quality patient care. Conventional manual systems lead to communication delays and scheduling conflicts, among other issues. The “Online Hospital Management System” a complete digital solution intended to modernize and streamline hospital administration, is presented in this study. MySQL is used to provide safe, scalable and relational data management, and PHP, Laravel framework are used in the system’s development. The main objective is to automate key hospital operations, such as a robust role-based access control (RBAC) system for administrators, doctors, staff and patients, a dynamic appointment scheduling system with automated doctor consultation slot management, and a digital diagnostic services module for creating and gathering medical reports. The user can book their appointment without registration; in this project the appointment portal is publicly visible and after that user get a notification via email that the appointment is confirmed/reject. Also, the patient get also reminder before 1 hour of appointment time. In this project each doctor has 20 patient booking system, when 20 patient booking is done others cannot take appointment. Using an agile methodology, the development process improved user interfaces and workflows by incorporating iterative feedback from medical staff. Testing confirms that the system reduces admin work and avoid scheduling conflicts or errors. Online platform available for patient 24/7 to improve patient care services. This project illustrates how this web-based automation increased efficiency lowers the number of cancellations. Overall, this platform supports long-term success by creating a paperless healthcare service.

Acknowledgements

First of all, I would like to thank the Almighty Allah for Her endless blessings, strength, and guidance, which enabled me to complete this practicum successfully.

I am deeply grateful to my respected supervisor, Jannatul Ferdous, for giving me the opportunity to complete this practicum under her supervision. Her continuous support, valuable guidance, motivation, and constructive feedback were extremely helpful throughout the preparation of this report. I am truly thankful for her patience and encouragement.

I am thankful to the Department of Computer Science and Engineering (CSE), IUBAT, for their support and guidance. Special thanks to Assistant Professor Shahinur Alam for his valuable feedback. I am also grateful to Dr. Utpal Kanti Das, Chairman of the Department, and all respected faculty members for their encouragement and inspiration.

Finally, I would like to thank my parents, family members, friends, and well-wishers for their constant encouragement, moral support, and inspiration, which played a vital role in helping me complete this practicum successfully.

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Chapter 1.

Introduction

In the current healthcare landscape, many hospitals still depend on manual processes that cause errors such as doctor double-booking and misplaced patient reports, ultimately affecting patient care. To overcome these issues, we developed an Online Hospital Management System, a unified platform that digitizes and automates daily hospital operations. The system enables patients to book appointments online by viewing doctors' real-time availability, while a logic-based scheduling engine limits each doctor to 20 appointments per day to prevent overbooking. Automated email reminders and panel notifications help reduce missed appointments. On the administrative side, admin manage staff and approvals, doctors access their daily schedules and check how many patients take their appointments, and lab technicians upload diagnostic reports that patients can securely download online. Overall, the system provides a secure, efficient, and accessible healthcare solution, connecting all system users through a single web-based platform available 24/7.

1.1 Background of the Study

In the modern healthcare system are essential for saving lives and ensuring patient satisfaction. Traditional hospital management system commonly depends on manual documentation and Non-integrated systems, increasing the problems in patient records, scheduling conflicts, and long waiting times. In modern technology advances, digital systems are increasingly needed to centralize information, automate routine tasks, and facilitate smooth communication between patients, doctors, and administrative staff. The "Integrated Web-Based Hospital Management System for Automated Services" was developed to address these challenges by utilizing web technologies to create an integrated platform for healthcare administration.

1.2 Methodology

The development of the "Online Hospital Management System" was grounded in a thorough research process aimed at understanding the specific operational challenges of healthcare institutions. To ensure the system was both practical and robust, data was gathered from two distinct categories of information sources.

1.2.1 Primary Sources

We got the information for this project by talking to people and watching what they do. This meant we had conversations with people who work in healthcare like the people, in the office and the doctors so we could see what they do every day and what problems they have like it is hard to keep track of appointments in a book and people not showing up. We also watched how things are done now to see where things get stuck, like how it takes to sign people in and get their medical records from the files. These direct insights were crucial in defining the core requirements of our system, specifically the need for the "20-patient daily limit" per doctor to ensure quality care and the automated email reminder system to reduce missed appointments.

1.2.2 Secondary Sources

To help with the results a lot of extra work was done by looking at other sources. This meant going over the papers for the Laravel framework and the MySQL database to make sure that the coding and security methods used are what most people in the industry use. Papers from schools and online studies about Hospital Management Information Systems were also looked at to see how to make the User Interface (UI) easy to use for staff who are not good, with technology. Furthermore, existing open-source healthcare platforms were evaluated to benchmark our features against current market standards, guiding the decision to implement a responsive, web-based architecture that allows for remote patient access.

1.3 Objectives

The main goals of the proposed system are split into two parts: the picture and the details. The proposed system has two kinds of objectives: Broad objectives and Specific objectives. These objectives of the proposed system are as follows:

1.3.1 Broad Objective

The main goal of this practicum work is to create an online Hospital Management System. This Hospital Management System is supposed to automate the tasks that the hospital staff has to do every day. It will help the hospital to move away from the way of doing things on paper. The Hospital Management System will make the hospital work in a digital way. This will make things easier for the patients. The Hospital Management System is, about making the hospital a paperless place where everything is done digitally and the patients get the best care.

1.3.2 Specific Objectives

a. To create a secure, role-based platform that allows hospital administrators to manage doctor profiles, staff rosters, and departmental data from a single interface.

b. To build a robust booking engine that prevents overbooking by strictly enforcing a daily limit of 20 appointments per doctor and managing real-time slot availability.

c. To integrate an automated notification service that sends email reminders to patients one hour prior to their scheduled appointments, thereby reducing no-show rates.

d. To provide a secure patient portal for the 24/7 retrieval of digital medical reports and test results, eliminating the need for physical hospital visits to collect paperwork.

e. To design a user-friendly public frontend that allows unregistered guests to easily view doctor schedules and request appointments online without complex registration hurdles.

1.4 Process Model

For the development of the " Online Hospital Management System," the Agile Process Model was utilized. Agile is an iterative approach to software development that emphasizes flexibility, customer collaboration, and the delivery of small, functional pieces of the software in rapid cycles known as "sprints." Without waiting to complete the entire system at once, the project was broken down into manageable modules (Appointment Module, Report Module, Admin Dashboard), which were developed, tested, and refined independently. This model allowed for continuous evaluation and the ability to adapt to new requirements such as adding the "20-patient limit" or "1-hour automated reminders" even late in the development phase. Reasons for Choosing the Agile Model:

a. **Adaptability:** Agile allowed quick changes, such as adding daily appointment limits, without affecting overall development.

b. **Early Bug Detection:** Module-wise testing helped identify and fix issues early.

c. **Iterative Delivery:** A working prototype was delivered early, starting with patient registration.

d. **User-Centric Design:** Continuous feedback improved usability for doctors and patients.

e. **Risk Reduction:** Breaking the system into smaller modules minimized overall project risk.

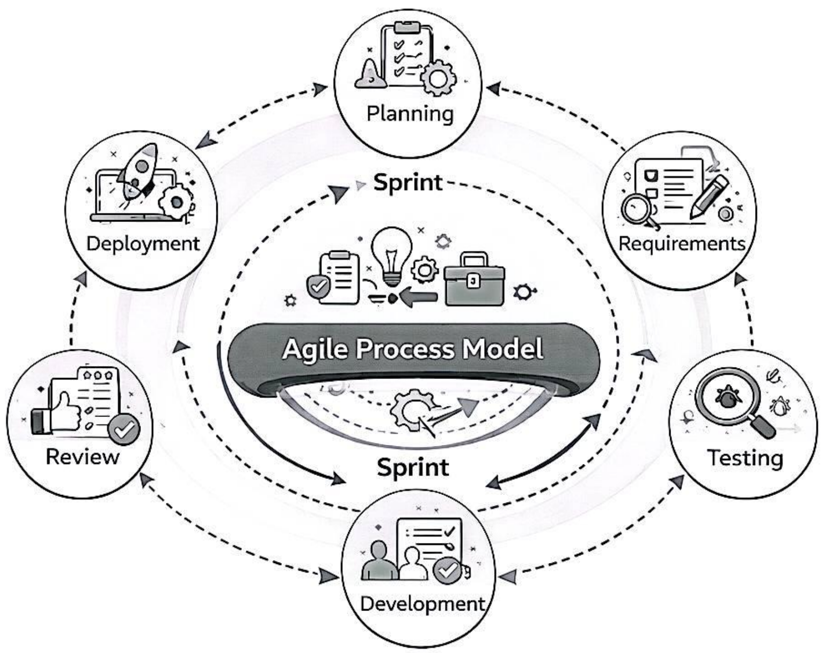


Figure 1.1 Agile Process Model

1.5 Feasibility Study

The feasibility study has been divided into three phases: technical feasibility, economic feasibility, and operational feasibility. The proposed system is feasible in all these three phases, as discussed below. This study helps to evaluate whether the system can be developed efficiently within the available resources, budget, and organizational environment. It also ensures that the system will be practical, cost-effective, and acceptable to end users before full-scale implementation.

1.5.1 Technical Feasibility

The project is technically feasible as it utilizes standard, mature, and widely supported technologies that ensure reliability and scalability. The system is developed using Laravel, a robust PHP framework known for its stability, strong security features, and ability to handle complex data relationships and high traffic loads common in hospital environments. For data management, MySQL is used as a reliable relational database management system to securely store and efficiently retrieve large volumes of patient, doctor, and medical records. As a web-based application developed using HTML5, CSS3, and JavaScript, the system provides cross-platform accessibility through any standard web browser without the need for high-end hardware. Additionally, Laravel’s built-in security mechanisms, including CSRF protection, secure authentication, and authorization policies, ensure the protection of sensitive medical data and support compliance with data privacy standards.

1.5.2 Economic Feasibility

The project is economically feasible because the development and operational costs are relatively low compared to the long-term benefits and cost savings it provides. The system is developed using open-source technologies such as PHP, Laravel, and MySQL, which significantly reduce software licensing and acquisition costs. By automating manual processes including appointment scheduling, report generation, and billing, the hospital can reduce administrative overhead and minimize expenses related to paperwork and stationery. Furthermore, the system improves operational efficiency by reducing human errors and accelerating key processes such as patient record retrieval and billing. This enables hospital staff and doctors to serve more patients within the same time frame, which can potentially increase overall productivity and revenue in the long run.

1.5.3 Operational Feasibility

The project is operationally feasible as it effectively addresses existing inefficiencies in manual hospital management while being designed for ease of use and practical adoption. The system follows a user-centric design approach by providing distinct and intuitive dashboards for different roles such as Admin, Doctor, Staff, and Patient. This role-based access ensures that users interact only with features relevant to their responsibilities, thereby reducing complexity and minimizing the learning curve. Additionally, the system resolves critical operational issues such as appointment clashes, misplaced medical records, and scheduling conflicts through automated logic and centralized data management. The workflow of the system closely aligns with real-world hospital operations, from patient appointment booking to prescription generation and billing, enabling smooth integration into daily activities without disrupting existing procedures.

Chapter 2.

Organizational Overview

2.1 Organization Overview

GenNet is a well-established organization operating as a sister concern of Rahimafrooz Group, one of the most reputed and diversified conglomerates in Bangladesh. As part of the Rahimafrooz family, GenNet upholds the group's long-standing commitment to quality, innovation, and customer-centric solutions. The organization benefits from the strong corporate governance, industry experience, and ethical standards that Rahimafrooz has maintained for decades. GenNet focuses on delivering technology-driven and network-based solutions that support modern business operations. The organization is involved in providing IT infrastructure, networking services, and digital solutions aimed at enhancing operational efficiency and reliability for its clients. By leveraging advanced technologies and skilled professionals, GenNet ensures dependable service delivery and continuous improvement. As a sister concern of Rahimafrooz, GenNet plays a strategic role in supporting the group's digital transformation initiatives while also serving external clients across various sectors. The organization emphasizes professionalism, innovation, and sustainability, aligning its goals with the broader vision of Rahimafrooz Group to contribute positively to national development and technological advancement.

2.2 Organization Vision

The vision of GenNet is to become a leading provider of innovative and reliable technology and networking solutions in Bangladesh by enabling digital transformation, operational efficiency, and sustainable growth for businesses. The organization aims to contribute to national development through the use of modern technologies while maintaining the highest standards of quality, integrity, and customer satisfaction.

2.3 Organization Mission

The mission of GenNet is to deliver secure, scalable, and cost-effective IT and network solutions that meet the evolving needs of its clients. The organization is committed to leveraging skilled professionals, advanced technologies, and best industry practices to ensure dependable service delivery. As a sister concern of Rahimafrooz Group, GenNet strives to uphold excellence, ethical business practices, and continuous innovation in all its operations.

2.4 Organization Services

GenNet provides a wide range of technology-based services aimed at supporting modern business operations. The key services offered by the organization include:

a. IT infrastructure and network solutions
b. System integration and technical support services
c. Web-based and software solution development
d. Network security and data management solutions
e. Maintenance and support services for IT systems
f. Consultancy services for digital transformation

2.5 Organizational Structure

The organizational structure of GenNet follows a hierarchical and functional model to ensure efficient management and clear communication. The organogram of the organization is illustrated in Figure 2.1, which shows the distribution of responsibilities and reporting relationships among top management, departmental heads, technical teams, and support staff. This structured approach enables smooth coordination, accountability, and effective decision-making within the organization.

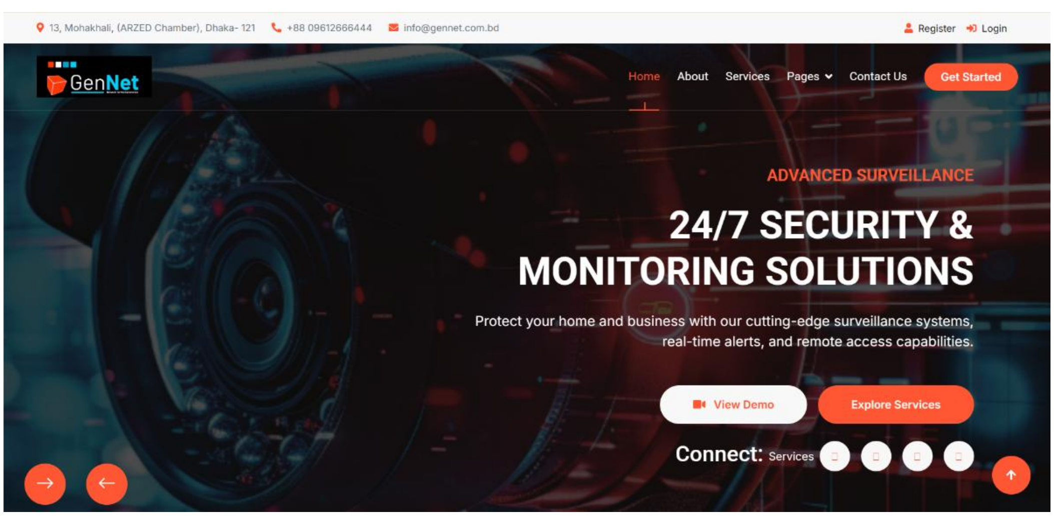


Figure 2.1 Organizational Structure

2.6 My Position in this Organization

During my internship period at GENNET, a sister concern of Rahimafrooz Group, I worked as an Intern Software Engineer. In this role, I was involved in supporting the software development team in designing, developing, and maintaining web-based applications. My responsibilities included assisting in frontend and backend development tasks, debugging and testing application features, and understanding system workflows under the guidance of senior software engineers. As an Intern Software Engineer, I gained practical exposure to real-world software development practices, including version control, database handling, and framework-based development. This position allowed me to apply theoretical knowledge acquired from academic coursework to practical projects while developing professional skills such as teamwork, problem-solving, and effective communication within a corporate environment.

2.7 Address of the Organization

13, Mohakhali, (ARZED Chamber), Dhaka- 1212

Map: https://maps.app.goo.gl/sRYAEeVJNE9y36a59

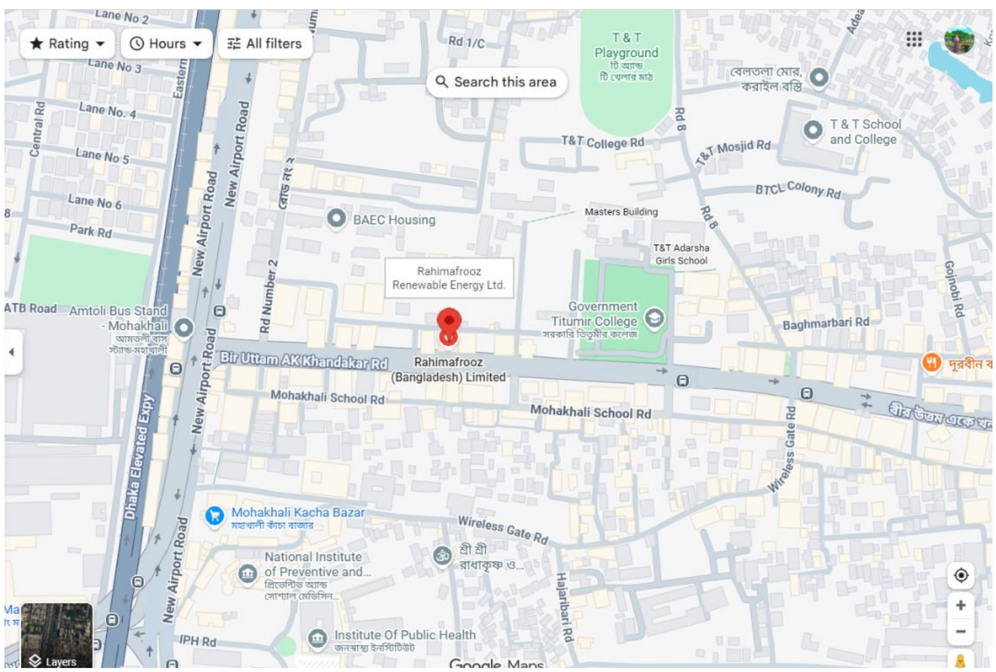


Figure 2.2 Organizational Location Map

Chapter 3.

Requirement Engineering

3.1 Requirement Engineering

Requirement Engineering is the disciplined application of proven principles, methods, and tools to describe the behavior of the Hospital Management System and its associated constraints. This phase focused on understanding the actual needs of the stakeholders (Administrators, Doctors, Staff, and Patients) to ensure the final product effectively solves their operational challenges. The process was carried out in the following stages:

3.1.1 Requirement Elicitation

This step involved gathering raw requirements from various sources to understand what the system needed to do. The methods used included:

- **Stakeholder Analysis:** Identifying the key users of the system (Doctors, Patients, Admin, Receptionists) and understanding their specific privileges and workflows.
- **Observation:** Observing the manual processes currently used in hospitals, such as paper-based appointment booking and physical file storage, to identify bottlenecks and inefficiencies.
- **Analysis of Similar Systems:** Reviewing existing hospital management software to identify standard features and potential areas for improvement, such as better user interfaces or more secure data handling.

3.1.2 Requirement Analysis

After gathering the requirements, they were analyzed to identify inconsistencies, ambiguities, and feasibility.

- **Classification:** Requirements were categorized into Functional Requirements (System must generate a PDF report) and Non-Functional Requirements (System must load pages under 2 seconds).
- **Prioritization:** Core features like User Authentication, Appointment Scheduling, and Billing were prioritized over secondary features to ensuring the development of a Minimum Viable Product (MVP) first.

- **Conflict Resolution:** Potential conflicts, such as the need for detailed data vs. patient privacy, were resolved by designing a robust role-based access control (RBAC) system.

3.1.3 Requirement Specification

The analyzed requirements were formally documented.

- This resulted in a comprehensive set of specifications detailing how the Laravel backend should handle logic (Controllers, Models) and how the frontend (Blade Templates) should present information.
- Specific workflows were mapped out, such as the Appointment Cycle (Request -> Approval -> Visit -> Prescription) and the Billing Cycle (Test Request -> Payment -> Receipt).

3.1.4 Requirement Validation

The final stage involved ensuring the documented requirements accurately reflected the user needs. This was done by verifying that the proposed modules (Admin Dashboard, Doctor Panel, Patient Portal) technically aligned with the operational goals of the hospital.

3.2 Requirement Engineering

This section defines the specific behavior and constraints of the system, categorized into functional requirements (what the system does) and non-functional requirements (how the system performs).

3.2.1 Functional Requirements

These requirements define the specific functions and interactions within the system for different user roles.

1. User Authentication & Authorization

a. The system allows users to register (Patients) and log in.

b. The system must restrict access based on roles: Admin, Doctor, Staff, and Patient.

c. Users must be able to update their profiles and change passwords securely.

2. Admin Module

a. User Management: Admin can add, update, and delete Doctors and Staff members.

b. Department Management: Admin can manage hospital departments (Cardiology, Neurology, etc.).

c. Scheduling: Admin manages doctor schedules and staff rosters.

d. Reporting: Admin can generate and view system-wide activity reports (appointments, revenue).

3. Doctor Module

a. Appointment Management: Doctors can view their daily and upcoming appointment lists.

b. Patient Care: Doctors can add diagnosis notes and writing prescriptions for patients.

c. History: Doctors can view the treatment history of specific patients.

d. 4. Patient Module

e. Booking: Patients can browse doctors by department and request appointments for specific dates and times.

f. Dashboard: Patients can view the status of their appointments (Pending, Confirmed, Completed).

g. Medical Records: Patients can view and download their medical reports and prescriptions.

5. Staff Module

a. Billing: Staff can create invoices and process payments for tests.

b. Test Requests: Staff handles laboratory test requests and manages their status.

c. Queue Management: Staff oversees the daily patient queue and check-in process.

3.2.2 Non-Functional Requirements

These requirements define the quality attributes, performance constraints, and security measures of the system.

1. Security

a. Data Protection: Passwords must be hashed (Bcrypt) before storage in the database.

b. Access Control: Critical routes must be protected by middleware to prevent unauthorized access (a patient cannot access admin routes).

c. CSRF Protection: All forms must include tokens to prevent Cross-Site Request Forgery attacks.

2. Performance & Efficiency

a. Response Time: The system should load web pages within 2-3 seconds under normal network conditions.

b. Concurrency: The system must handle multiple users (doctors, staff, patients) accessing the database simultaneously without data corruption.

3. Usability

a. Interface: The user interface must be clean, professional, and intuitive, requiring minimal training for staff and doctors.

b. Responsiveness: The application must be responsive, adapting its layout for desktops, tablets, and mobile devices.

4. Reliability & Availability

a. Uptime: The system should be available 24/7 to allow patients to book appointments at any time.

b. Data Integrity: The database must ensure referential integrity (e.g., an appointment cannot exist without a valid patient and doctor).

3.3 Use Case Diagram

Figure 3.1 Use Case Diagram.

Chapter 4.

Analysis

4.1 Activity Diagram

The activity diagram is a graphic representation of the hospital management systems operational processes, also the cycle for scheduling appointments and billing. It highlights the transfer of control between the user and the system by breaking these interactions down into discrete, sequential phases.

4.1.1 Patient Appointment Booking Activity Diagram

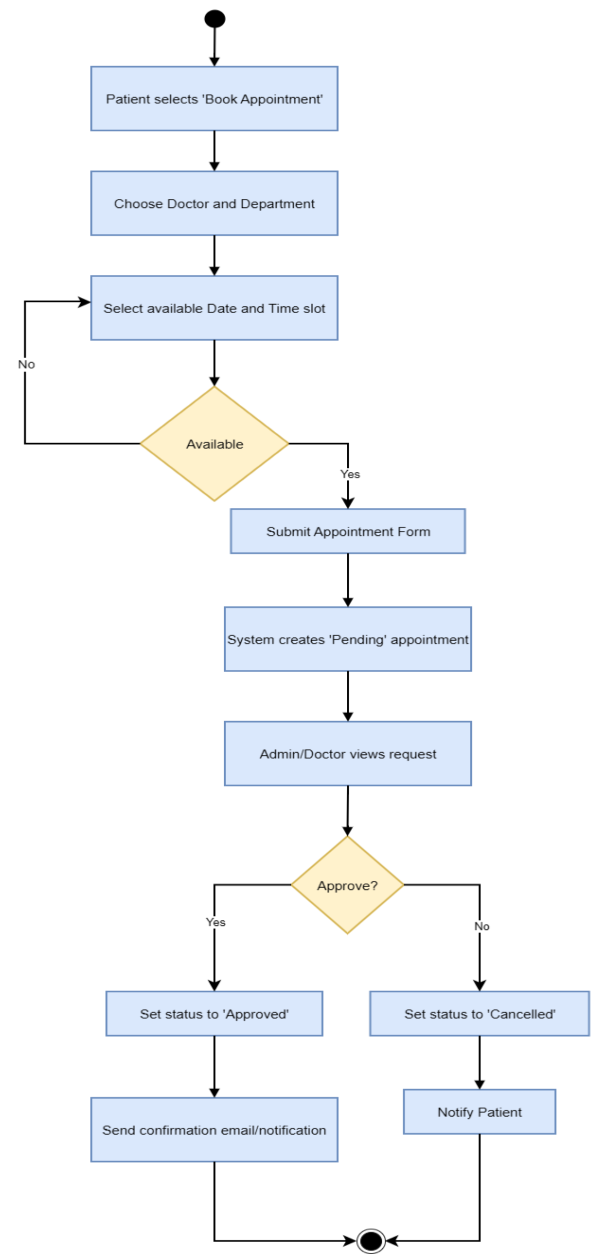


Figure 4.1 Patient Appointment Booking Activity Diagram

4.1.2 Doctor/Staff Leave Application Activity Diagram

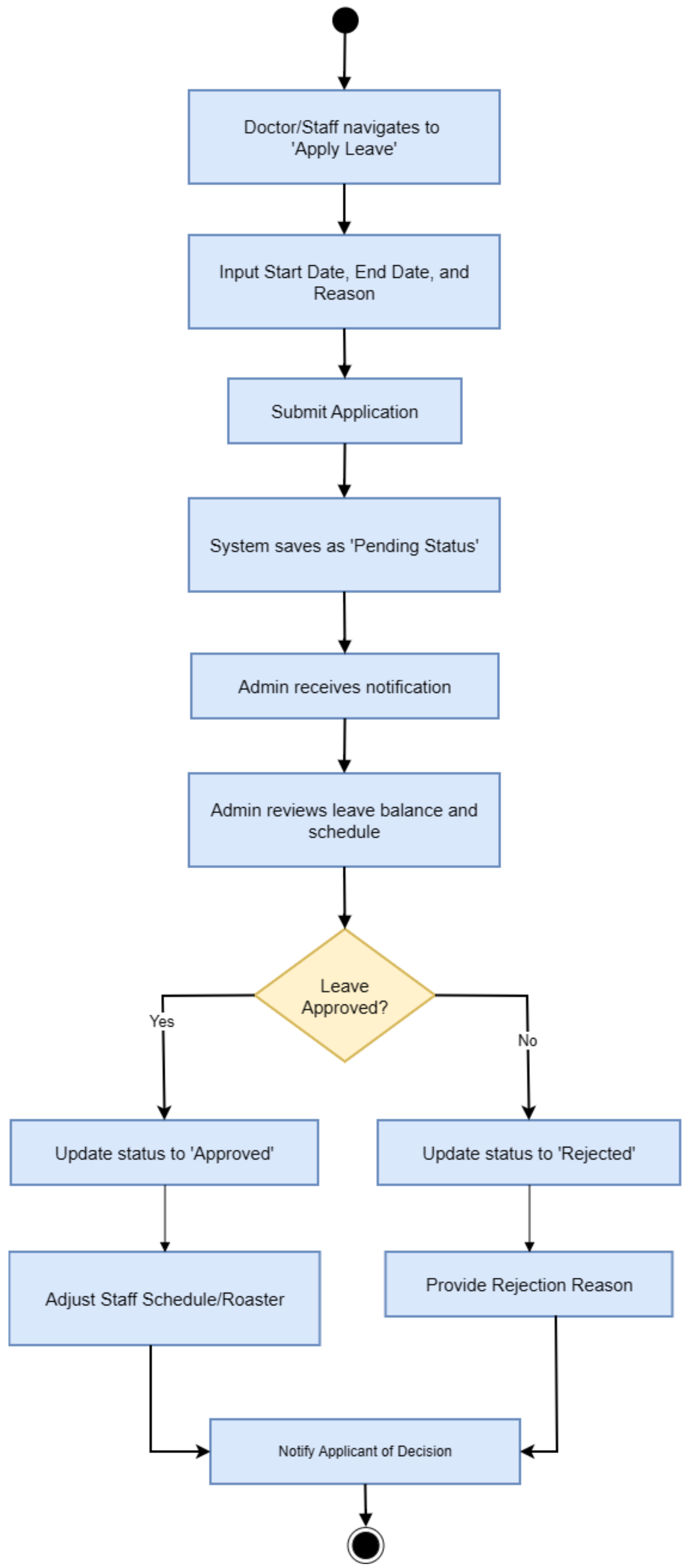


Figure 4.2 Doctor/Staff Leave Application Activity Diagram

4.1.3 Admin Staff Task Management Activity Diagram

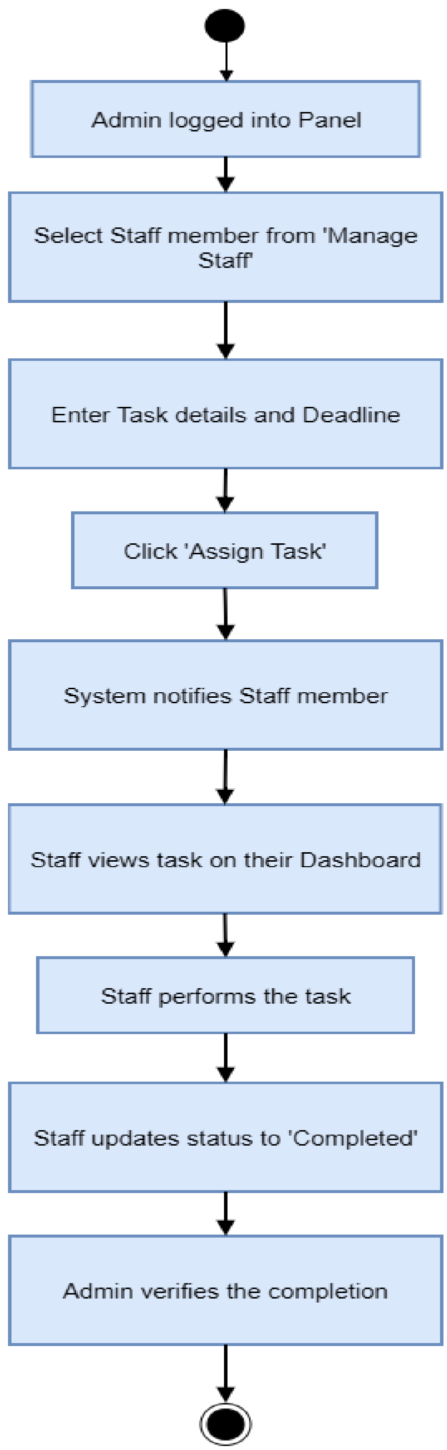


Figure 4.3 Admin Staff Task Management Activity Diagram

4.2 Swimlane Diagram

The systems role-based tasks are visualized using a swim lane diagram. Each stockholder's obligations and interactions are depicted. This project provides a clear example of how hospital operations, including appointment scheduling, consultations, billing and report management, are coordinated by patients, administrators, doctors and staff.

4.2.1 Patient Appointment Booking (Swimlane)

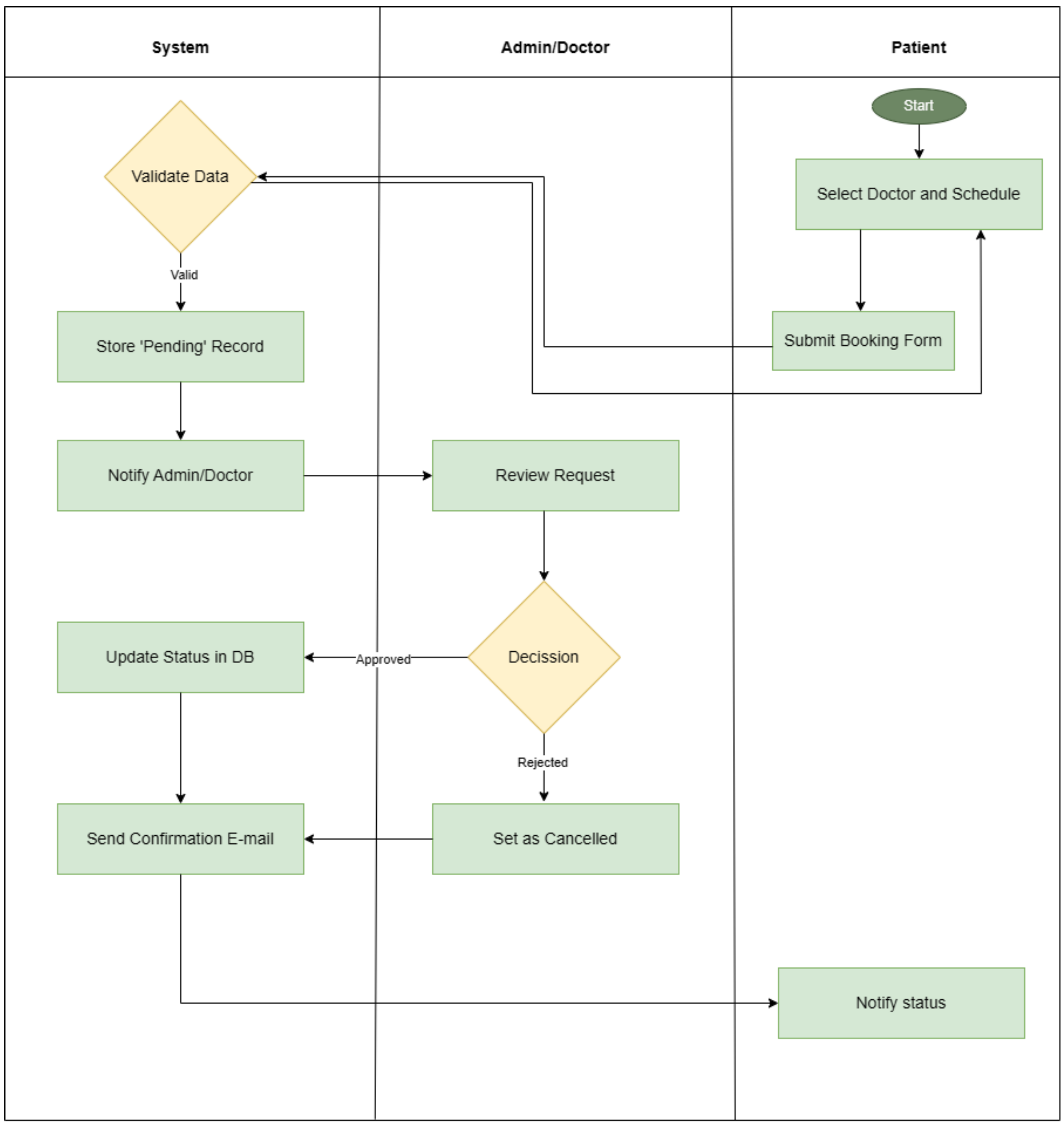


Figure 4.4 Patient Appointment Booking (Swimlane)

4.2.2 Staff Leave Application (Swimlane)

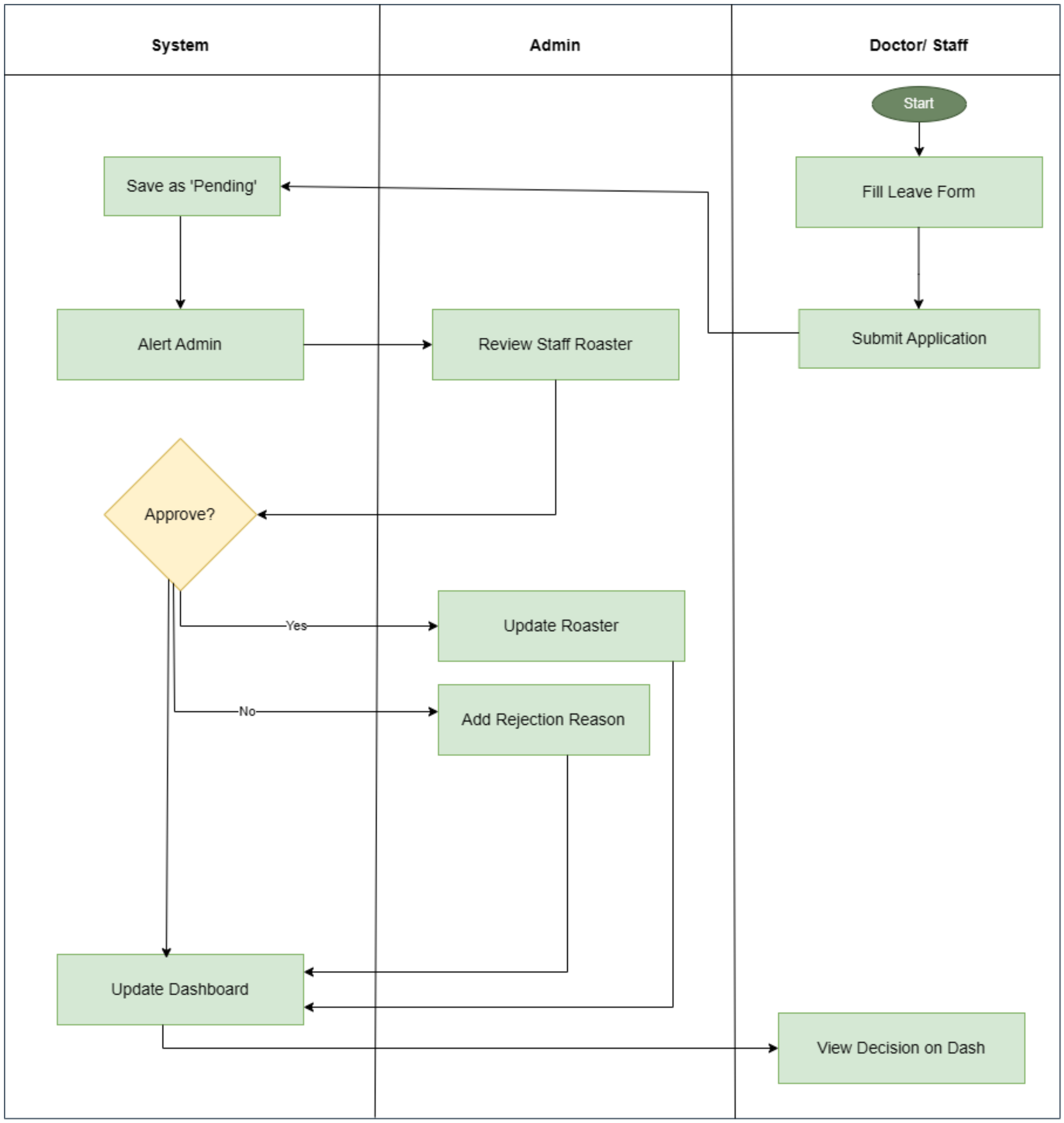


Figure 4.5 Staff Leave Application (Swimlane)

4.2.3 Staff Task Management (Swimlane)

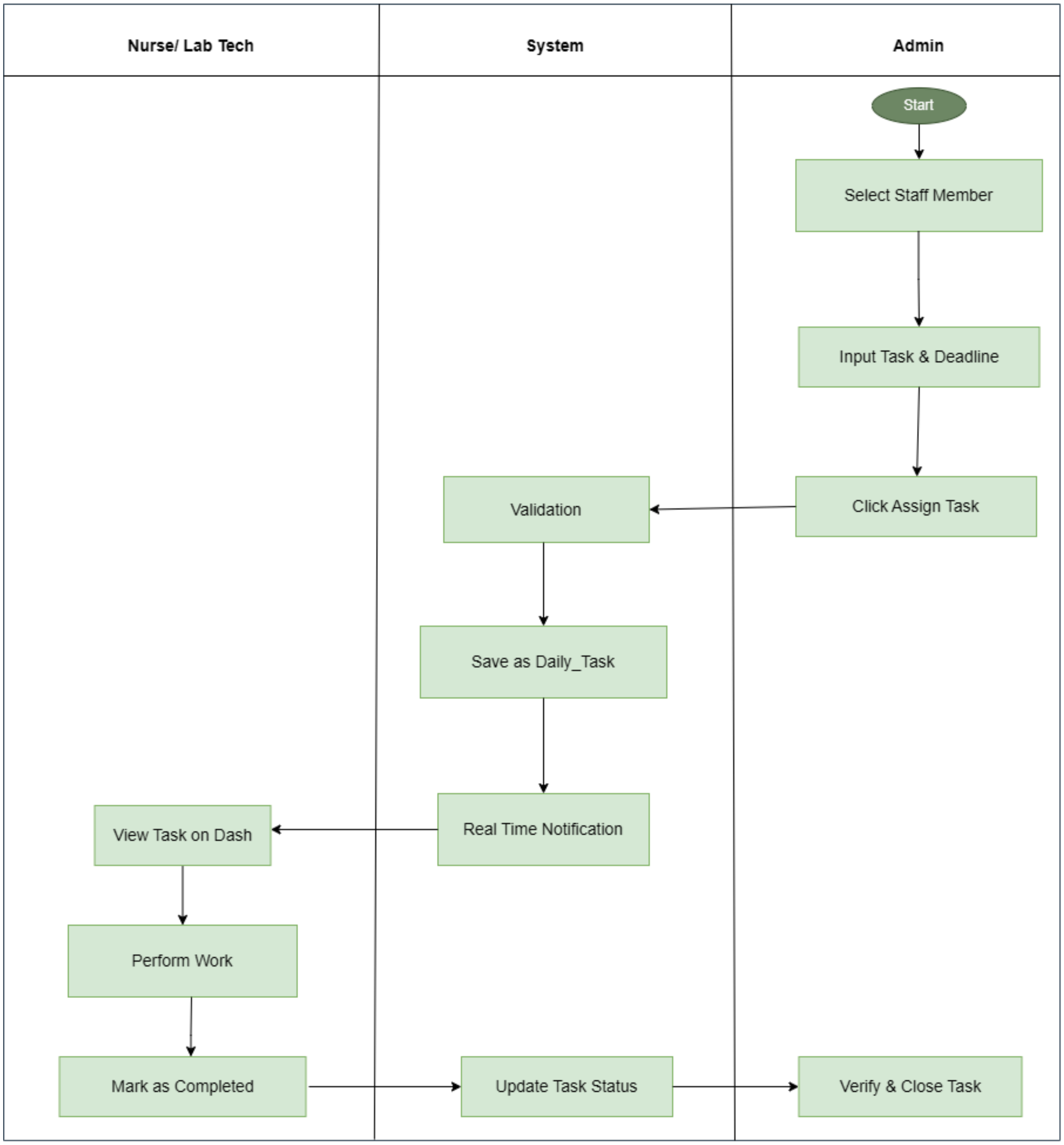


Figure 4.6 Staff Task Management (Swimlane)

4.3 Class Diagram

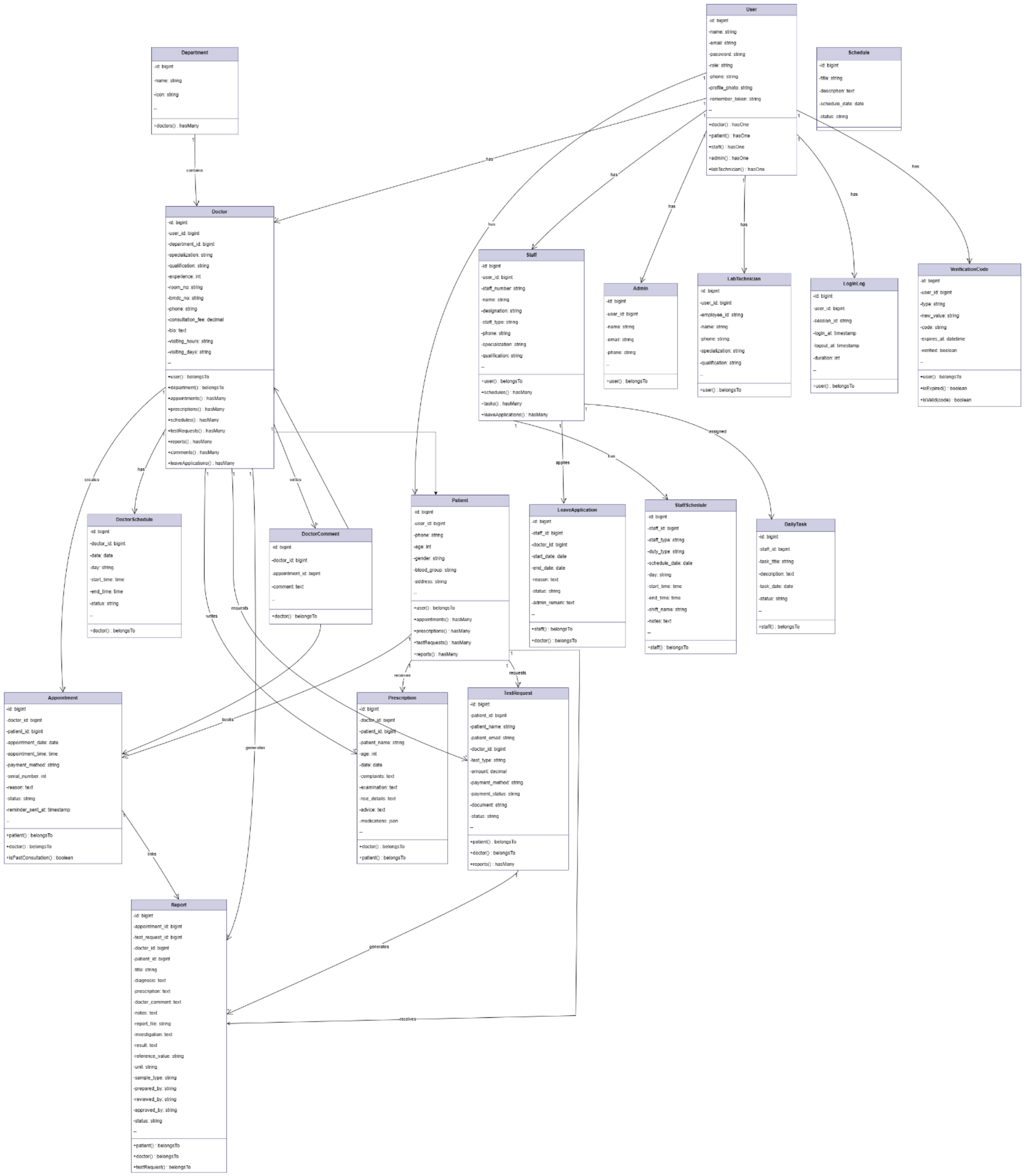


Figure 4.7 Class Diagram

4.4 Sequence Diagram

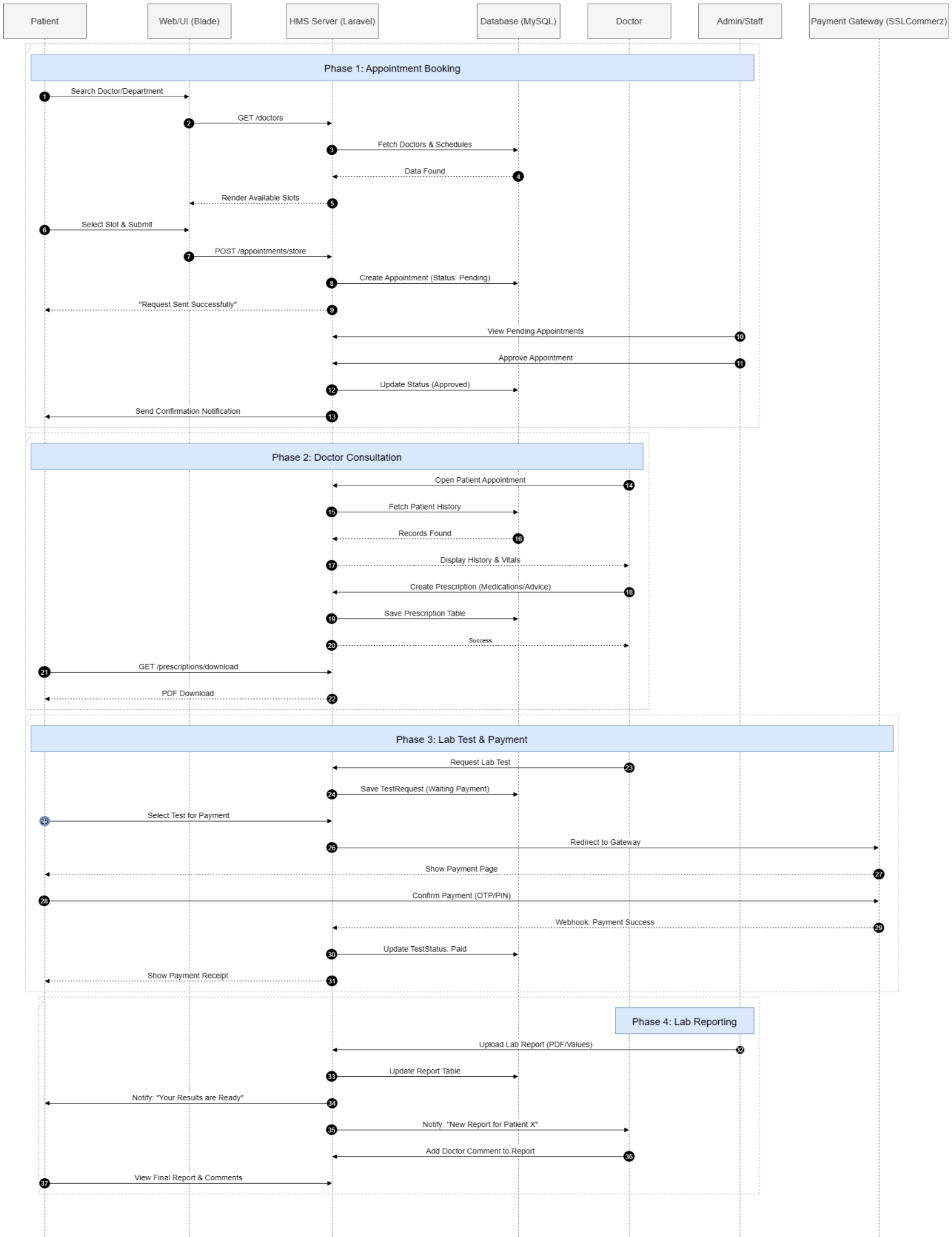


Figure 4.8 Sequence Diagram

4.5 CRC Cards

CRC (Class-Responsibility-Collaborator) cards is used to identify and organize classes of a system in object-oriented design. Each card represents a class and lists its responsibilities and its collaborators.

Table 4.1 CRC of Patient

<table>
<tr><td colspan="2" align="center">Class: Patient</td></tr>
<tr><td>Responsibilities:
Store patient personal information
Manage patient medical history
Book appointments with doctors
View appointment status
Download medical reports and prescriptions
Make payments for services</td><td>Collaborators:
User (inherits authentication)
Appointment (creates and views)
Doctor (selects for appointment)
Report (views and downloads)
Test Request (requests and pays for)
Department (browses doctors by)</td></tr>
</table>

Table 4.2 CRC of Doctor

<table>
<tr><td colspan="2" align="center">Class: Doctor</td></tr>
<tr><td>Responsibilities:
Store doctor professional information
Manage consultation schedule
View assigned appointments
Conduct patient consultations
Create prescriptions and diagnoses
Request laboratory tests
View patient medical history</td><td>Collaborators:
User (inherits authentication)
Department (belongs to)
Appointment (receives and completes)
Patient (treats)
Report (creates medical reports)
Doctor Schedule (defines availability)
Test Request (orders tests)</td></tr>
</table>

Table 4.3 CRC of Staff

Class: Staff	
Responsibilities: • Manage patient queue and check-in • Process test requests • Generate invoices and receipts • Collect and confirm payments • Upload laboratory test results • Manage work schedules and shifts	**Collaborators:** • User (inherits authentication) • Department (assigned to) • Patient (serves) • Test Request (creates and processes) • Report (uploads test results) • Staff Schedule (follows duty roster) • Payment Gateway (processes payments)

Table 4.4 CRC of Admin

Class: Admin	
Responsibilities: • Manage system users (doctors, staff) • Approve or reject appointment requests • Manage hospital departments • Create and assign schedules • Generate system-wide reports • Monitor user activity and login logs • Configure system settings	**Collaborators:** • User (inherits authentication) • Doctor (creates and manages) • Staff (creates and manages) • Department (creates and manages) • Appointment (approves/rejects) • Doctor Schedule (creates and assigns) • Staff Schedule (creates rosters) • Login Log (monitors activity) • Email Service (sends notifications)

Table 4.5 CRC of Appointment

Class: Appointment	
Responsibilities: • Store appointment details (date, time, status) • Track appointment lifecycle (pending → confirmed → done) • Validate slot availability • Send reminder notifications • Link to medical reports	**Collaborators:** • Patient (belongs to) • Doctor (assigned to) • Admin (approved by) • Doctor Schedule (checks availability) • Report (generates after completion) • Email Service (sends notifications)

Table 4.6 CRC of Report

Class: Report	
Responsibilities: • Store medical diagnosis and prescriptions • Store laboratory test results • Manage report file uploads (PDF/images) • Generate downloadable reports • Track report status	**Collaborators:** • Patient (belongs to) • Doctor (created by) • Appointment (linked to) • Test Request (linked to) • PDF service (generates PDF) • Email Service (notifies patient)

Table 4.7 CRC of Test Request

Class: Test Request	
Responsibilities: • Store test order information • Calculate test costs • Track payment status • Manage test result uploads • Generate invoices and receipts	**Collaborators:** • Patient (requested by) • Doctor (prescribed by) • Staff (processed by) • Report (generates result report) • Payment Gateway (processes payment) • PDF Service (generates receipt)

Table 4.8 CRC of Department

Class: Department	
Responsibilities: • Store department information • Organize doctors by specialization • Organize staff by function • Display department services	**Collaborators:** • Doctor (contains multiple) • Staff (contains multiple) • Patient (browsed by) • Admin (managed by)

Table 4.9 CRC of Doctor Schedule

Class: Doctor Schedule	
Responsibilities: • Define doctor availability (days, times) • Calculate available time slots • Block unavailable dates • Prevent double-booking	**Collaborators:** • Doctor (belongs to) • Appointment (validates against) • Admin (created/modified by)

Chapter 5.

Risk Management

5.1 Risk Management

Risk management is a process of identifying, analyzing, and addressing issues that can negatively affect a project's success. It helps team to prepare potential risk and how to minimize them. Risk planning can mitigate risk of a system and make it efficient.

Stages of Risk Management-

a. Risk Identification
b. Risk Analysis
c. Planning
d. Monitoring

5.2 Risk Identification

Table 5.1 Risk Identification

Risk Type	Possible Risk
Technical	System crashes, Laravel/PHP/MySQL issues
Security	Data breaches, unauthorized access to patient records
Financial	Payment gateway failures (bKash, Nagad, Rocket, etc.)
Operational	Staff misuse, data entry errors, appointment delays
Legal/Compliance	HIPAA, GDPR, medical data laws
Project Management	Delays, scope creep, resource issues
User Experience	Poor UI, slow response, mobile issues
Integration	Email, SMS, payment API failures
Data Management	Migration, backups, reporting issues
Performance	Concurrent users, peak hour loads, scalability

5.3 Risk Analysis

Table 5.2 Risk Analysis

Possible Risks	Probability	Impact
System Crashes/Server downtime	30-40% (Moderate)	Serious
Data breaches/Security	40-50% (High)	Catastrophic
Payment gateway failures	10-20% (Low)	Serious
Staff misuse/data errors	20-30% (Moderate)	Tolerable
Legal compliance failures	15-25% (Low)	Catastrophic
Project delays/scope creep	30-40% (Moderate)	Serious
Poor UX/performance	20-30% (Moderate)	Serious
Integration failures	15-25% (Low)	Serious
Data migration issues	25-35% (Moderate)	Serious
Scalability/concurrency	30-40% (Moderate)	Serious

5.4 Risk Planning

Table 5.3 Risk Planning

Risk	Strategy
System downtime / crashes	Cloud hosting that includes Laravel queue workers, load balancing, automated backups, and monitoring tools.
Security breach / hacking	SSL/HTTPS, RBAC, @FA, CSRF protection, bcrypt hashing and security assessments.
Payment gateway failure	Transaction logging, fallback techniques and several gateways (bkash, Nagad, Rocket, Mastercard, Visa).
Hospital staff misuse of platform	Role-specific rights, training, audit logs, authentication and authorization and usage guidelines.
Legal / compliance issues	Legal advice, data retention guidelines, consent management, and HIPAA/GDPR compliance.

Project delivery delay	Agile methodology, daily standups, milestones, project management tools and a 15-20% buffer.
Poor user experience / Low patient satisfaction	User Friendly interface, mobile responsiveness, 24/7 support, multiple channels of communication, usability testing.
Third-party integration failures	Sandbox testing, error handling, fallback options, API monitoring and documentation.
Data migration challenges	Test migrations, automated scripts, data mapping, validation and backups.
Performance / scalability issues	CDN, lazy loading, load testing, database indexing and caching (Redis/Memcached).

5.5 RMMM (Risk Mitigation Management and Monitoring) Planning

Throughout the system development lifecycle, possible risks are methodically identified, their impact is lessened, and they are continuously monitored using Risk mitigation, management and monitoring (RMMM) planning. The probability, impact, mitigation techniques, and management plans of the main project hazards are listed in the following tables.

Table 5.4 RMMM Planning for Risk 1

Project Risk 1	
Risk Name	System crashes, server downtime, or software bugs.
Probability	30-40% (Moderate)
Impact	Serious
Description	The system may go offline due to server overload, bugs or infrastructure failure, affecting user access and services.
Mitigation & Monitoring	Use cloud hosting, load balancers, automated backups and real time monitoring tools.
Management	IT team responds immediately and restores using backup recovery plans.
Status	Managed

Table 5.5 RMMM Planning for Risk 2

Project Risk 2	
Risk Name	Data breaches, hacking attempts, or unauthorized access
Probability	20% – 30% (Moderate)
Impact	Catastrophic
Description	Unauthorized access or data leakage may compromise sensitive patient and system data.
Mitigation & Monitoring	Implement SSL encryption, firewalls, role-based access control, and security audits.
Management	Security team applies patches, monitors logs, and handles incidents promptly.
Status	Managed

Table 5.6 RMMM Planning for Risk 3

Project Risk 3	
Risk Name	Payment gateway failure or transaction errors
Probability	10% – 20% (Low)
Impact	Serious
Description	Payment issues may prevent successful transactions or cause billing discrepancies.
Mitigation & Monitoring	Use reliable payment gateways, transaction validation, and error logging.
Management	Finance and development teams verify failed transactions and provide refunds if required.
Status	Managed

Table 5.7 RMMM Planning for Risk 4

Project Risk 4	
Risk Name	Staff misuse or data entry errors
Probability	20% – 30% (Moderate)

Impact	Tolerable
Description	Incorrect data entry or unauthorized actions by staff may affect data accuracy.
Mitigation & Monitoring	Provide staff training, validation rules, and activity logging.
Management	Admin monitors logs and corrects errors when detected.
Status	Managed

Table 5.8 RMMM Planning for Risk 5

Project Risk 5	
Risk Name	Legal or regulatory compliance failure
Probability	15% – 25% (Low)
Impact	Catastrophic
Description	Non-compliance with healthcare or data protection laws may cause legal penalties.
Mitigation & Monitoring	Follow data protection standards, maintain audit trails, and review policies regularly.
Management	Legal and compliance team ensures adherence to regulations.
Status	Managed

Table 5.9 RMMM Planning for Risk 6

Project Risk 6	
Risk Name	Project delays and scope creep
Probability	30%–40% (Moderate)
Impact	Serious
Description	Delays may occur due to changing requirements, extended features, or resource constraints.
Mitigation & Monitoring	Agile methodology, sprint planning, milestones, and regular stand-up meetings are followed.
Management	Project manager monitors timelines and controls scope changes.
Status	Managed

Table 5.10 RMMM Planning for Risk 7

Project Risk 7	
Risk Name	Poor user experience or performance issues
Probability	20%–30% (Moderate)
Impact	Serious
Description	Slow performance or complex UI may reduce user satisfaction and system adoption.
Mitigation & Monitoring	UI/UX optimization, performance testing, caching, and feedback collection are applied.
Management	Development team improves interfaces based on user feedback.
Status	Managed

Table 5.11 RMMM Planning for Risk 8

Project Risk 8	
Risk Name	Integration failures
Probability	15%–25% (Low)
Impact	Serious
Description	API testing, error handling mechanisms, and backup integration options are used.
Mitigation & Monitoring	Technical team troubleshoots and resolves integration issues.
Management	Development team improves interfaces based on user feedback.
Status	Managed

Table 5.12 RMMM Planning for Risk 9

Project Risk 9	
Risk Name	Data migration and backup issues
Probability	25%–35% (Moderate)
Impact	Serious

Description	Errors during data migration or backup processes may result in data loss or inconsistency.
Mitigation & Monitoring	Scheduled backups, migration testing, and rollback strategies are implemented.
Management	Database administrator validates data integrity regularly.
Status	Managed

Table 5.13 RMMM Planning for Risk 10

Project Risk 10	
Risk Name	Scalability and concurrency issues
Probability	30%–40% (Moderate)
Impact	Serious
Description	High user load may affect system responsiveness and stability.
Mitigation & Monitoring	CDN usage, load balancing, optimized queries, and scalable cloud resources are applied.
Management	System performance is continuously monitored and scaled when required.
Status	Managed

Chapter 6.

Project Planning and Scheduling

6.1 Project planning and scheduling

Project planning and scheduling are essential to the management of software development operations. Project planning is need to establishing the project's objectives, resources, tasks, hazards, and scope. It assures that it is clear what must be done and how. Project scheduling, on the other hand, arranges these tasks into a timeline that outlines when and how they should be finished. They both assist the development team in completing the project by the deadline.

6.2 Function Point Estimation

Function Point Analysis (FPA) is a standardized method for measuring the functional size of software applications. For the Hospital Management System, we will calculate the function points to estimate the project size, effort, and cost.

6.2.1. Functionality, Input, Output

The below table shows the functionality, input, and output of each function.

Table 6.1 Functionality, Input and Output.

Functionality	Input	Output
Patient Registration	Name, Email, Phone, Address, DOB, Gender, Blood Group, Emergency Contact	Patient account created, confirmation message or error
Doctor Registration	Name, Email, Phone, Specialization, Qualification, Experience, License Number, Department, Schedule	Doctor profile created, confirmation message
Staff Registration	Name, Email, Phone, Role, Department, Shift, Joining Date, Salary	Staff account created, confirmation message
User Authentication	Email, Password	Login successful, redirect to dashboard or error message
Appointment Booking	Patient ID, Doctor ID, Date, Time Slot, Request For, Symptoms, Payment Method	Appointment confirmed, receipt generated or error message
Doctor Schedule Management	Doctor ID, Day, Start Time, End Time, Max Patients, Status	Schedule updated, availability confirmation

Test Request Submission	Patient ID, Test Type, Description, Documents, Payment Method, Amount	Test request submitted, payment receipt or error
Medical Report Upload	Test Request ID, Report Title, Diagnosis, Notes, Report File	Report uploaded, patient notification sent
Payment Processing	User ID, Amount, Payment Method, Transaction ID, Gateway Response	Payment successful, receipt generated or transaction failed
Department Management	Department Name, Description, Icon, Head Doctor	Department created/updated, confirmation message
Leave Request	User ID, Leave Type, Start Date, End Date, Reason	Leave request submitted, pending approval notification
Appointment Cancellation	Appointment ID, Cancellation Reason	Appointment cancelled, refund processed or confirmation
Patient Search	Name, Phone, Patient Number	Patient list displayed with matching records
Doctor Search	Name, Specialization, Department	Doctor list with profiles and availability
View Appointment Details	Appointment ID	Complete appointment information, patient and doctor details
Check Doctor Availability	Doctor ID, Date	Available time slots, booking status
View Medical Report	Report ID	Report details, PDF download option, diagnosis and prescription
View Payment History	User ID	Transaction list, payment receipts, refund status
Staff Task Assignment	Staff ID, Task Title, Description, Deadline, Priority	Task assigned, notification sent to staff
Admin Dashboard	User activity logs, system statistics	Analytics, reports, user activity monitoring, system health
Email Notification	Recipient Email, Subject, Message, Appointment Details	Email sent, delivery confirmation

SMS Notification	Phone Number, Message, Appointment Reminder	SMS sent, delivery status
Generate Appointment Receipt	Appointment ID, Payment Details	PDF receipt with appointment and payment information
Generate Test Receipt	Test Request ID, Payment Details	PDF receipt with test details and payment confirmation
Generate Medical Report PDF	Report ID, Patient Details, Diagnosis	Formatted medical report PDF with hospital branding
User Profile Update	User ID, Updated Information	Profile updated, confirmation message
Password Change	Current Password, New Password, Confirm Password	Password updated successfully or validation error
View Staff Tasks	Staff ID	Assigned tasks list, deadlines, priorities, completion status
Department Statistics	Department ID	Patient count, doctor count, appointment statistics
User Activity Report	Date Range, User Role	Activity logs, login history, session duration, actions performed
Payment Reconciliation	Date Range, Payment Method	Transaction summary, total revenue, gateway-wise breakdown

6.2.2. Identify Complexity

Table 6.2 Identify Complexity of Data Function

Function Name	Types	Fields	DET	RET
Users	ILF	User ID, Name, Email, Password, Phone, Role, Email Verified, Phone Verified, Status, Created At, Updated At, Remember Token	12	1
Patients	ILF	Patient ID, User ID, Patient Number, DOB, Gender, Blood Group, Address, Emergency	13	1

		Contact, Emergency Phone, Medical History, Allergies, Current Medications, Insurance Info		
Doctors	ILF	Doctor ID, User ID, Doctor Number, Specialization, Qualification, Experience Years, License Number, Department ID, Consultation Fee, Visiting Days, Visiting Hours, Max Patients Per Day	12	1
Staff	ILF	Staff ID, User ID, Staff Number, Department ID, Position, Shift, Joining Date, Salary, Supervisor ID, Status	10	1
Departments	ILF	Department ID, Name, Description, Icon, Head Doctor ID, Status	6	1
Appointments	ILF	Appointment ID, Patient ID, Doctor ID, Appointment Date, Time Slot, Request For, Symptoms, Status, Payment Method, Payment Status, Amount, Notes	12	1
Test Requests	ILF	Test Request ID, Patient ID, Test Type, Description, Documents, Payment Method, Payment Status, Amount, Status, Admin Notes	10	1
Reports	ILF	Report ID, Test Request ID, Patient ID, Report Title, Diagnosis, Prescription, Doctor Comments, Admin Notes, Report File, Status	10	1
Payments	ILF	Payment ID, User ID, Payable Type, Payable ID, Amount, Payment Method, Transaction ID, Gateway Response, Status, Paid At	10	1
Doctor Schedules	ILF	Schedule ID, Doctor ID, Day of Week, Start Time, End Time, Max Patients, Is Available	7	1

Daily Tasks	ILF	Task ID, Staff ID, Assigned By, Task Title, Description, Priority, Status, Deadline	8	1
Leaves	ILF	Leave ID, User ID, Leave Type, Start Date, End Date, Total Days, Reason, Status, Approved By	9	1
Login Logs	ILF	Log ID, User ID, IP Address, User Agent, Login At, Logout At, Session Duration	7	1
Payment Gateway (bKash)	EIF	Transaction ID, Amount, Customer Phone, Status, Payment URL, Callback Data	6	1
Payment Gateway (Nagad)	EIF	Merchant ID, Order ID, Amount, Customer Phone, Payment Ref, Status	6	1
Payment Gateway (Rocket)	EIF	Transaction ID, Amount, Customer Account, Status, Reference	5	1
Email Service API	EIF	To, Subject, Body, Attachments, Status, Sent At	6	1
SMS Gateway API	EIF	Phone Number, Message, Status, Sent At, Delivery Status	5	1

Table 6.3 Identify Complexity of Transaction function

Function Name	Type	Fields	DET	FTR
Patient Registration	EI	Name, Email, Phone, Address, DOB, Gender, Blood Group, Emergency Contact, Password	9	2
Doctor Registration	EI	Name, Email, Phone, Specialization, Qualification, Experience, License Number, Department, Schedule, Password	10	3
Staff Registration	EI	Name, Email, Phone, Role, Department, Shift, Joining Date, Salary, Password	9	3

User Authentication	EI	Email, Password	2	2
Appointment Booking	EI	Patient ID, Doctor ID, Date, Time Slot, Request For, Symptoms, Payment Method, Amount	8	3
Doctor Schedule Update	EI	Doctor ID, Day, Start Time, End Time, Max Patients, Status	6	1
Test Request Submission	EI	Patient ID, Test Type, Description, Documents, Payment Method, Amount	6	2
Medical Report Upload	EI	Test Request ID, Report Title, Diagnosis, Notes, Report File, Status	6	1
Payment Processing	EI	User ID, Amount, Payment Method, Transaction ID, Gateway Response, Status	6	1
Department Creation	EI	Name, Description, Icon, Head Doctor, Status	5	1
Leave Request	EI	User ID, Leave Type, Start Date, End Date, Reason	5	2
Appointment Approval	EI	Appointment ID, Approved by Doctor, Approved by Admin	3	2
Leave Approval	EI	Leave ID, Approved by Admin	2	1
Appointment Cancellation	EI	Appointment ID, Cancellation Reason	2	1
Password Update	EI	Current Password, New Password, Confirm Password	3	1
Profile Update	EI	User ID, Name, Phone, Address, Other Details	5	2
Patient Dashboard	EO	Patient Info, Appointments, Reports, Test Requests, Payment History, Statistics	20	4
Doctor Dashboard	EO	Doctor Info, Schedule, Appointments, Patient List, Statistics	18	4

Admin Dashboard	EO	User Statistics, Appointment Analytics, Payment Summary, System Health, Activity Logs	25	10
Staff Dashboard	EO	Staff Info, Assigned Tasks, Schedules, Leave Status, Performance Metrics	15	4
Appointment Receipt PDF	EO	Appointment Details, Patient Info, Doctor Info, Payment Info, Hospital Branding	12	13
Test Request Receipt PDF	EO	Test Details, Patient Info, Payment Info, Hospital Branding	10	3
Medical Report PDF	EO	Report Details, Patient Info, Diagnosis, Prescription, Doctor Comments, Hospital Branding	15	4
User Activity Report	EO	User Info, Login Logs, Actions Performed, Session Duration, IP Address	12	3
Payment Reconciliation Report	EO	Transaction List, Total Revenue, Gateway-wise Breakdown, Date Range, Status Summary	14	3
Email Notification	EO	Recipient, Subject, Message, Appointment Details, Hospital Info	6	2
SMS Notification	EO	Phone Number, Message, Appointment Reminder	4	2
Search Patients	EQ	Search Query, Patient List with Name, Phone, Patient Number, DOB, Gender, Blood Group	7	1
Search Doctors	EQ	Search Query, Doctor List with Name, Specialization, Department, Qualification, Experience	6	2
View Appointment Details	EQ	Appointment ID, Complete Appointment Info, Patient Details, Doctor Details, Payment Status	12	3

Check Doctor Availability	EQ	Doctor ID, Date, Available Time Slots, Booking Status	4	3
View Medical Report	EQ	Report ID, Report Details, Diagnosis, Prescription, PDF Download Option	5	3
View Payment History	EQ	User ID, Transaction List, Payment Receipts, Refund Status	4	2
View Test Request Status	EQ	Test Request ID, Test Details, Status, Payment Info	4	2
View Staff Tasks	EQ	Staff ID, Task List, Deadlines, Priorities, Completion Status	5	2
View Department Details	EQ	Department ID, Department Info, Doctor List, Statistics	4	2
View User Profile	EQ	User ID, Profile Information	2	2

6.2.3 Unadjusted Function Point Contribution

Table 6.4 UFP of Transaction Function

Function Name	Type	DET	FTR	Complexity	UFP
Patient Registration	EI	9	2	Average	4
Doctor Registration	EI	10	3	High	6
Staff Registration	EI	9	3	High	6
User Authentication	EI	2	2	Low	3
Appointment Booking	EI	8	3	Average	4
Doctor Schedule Update	EI	6	1	Low	3
Test Request Submission	EI	6	2	Average	4
Medical Report Upload	EI	6	1	Low	3
Payment Processing	EI	6	1	Low	3
Department Creation	EI	5	1	Low	3
Leave Request	EI	5	2	Average	4
Appointment Approval	EI	3	2	Low	3

Leave Approval	EI	2	1	Low	3
Appointment Cancellation	EI	2	1	Low	3
Password Update	EI	3	1	Low	3
Profile Update	EI	5	2	Average	4
Patient Dashboard	EO	20	4	High	7
Doctor Dashboard	EO	18	4	High	7
Admin Dashboard	EO	25	10	High	7
Staff Dashboard	EO	15	4	High	7
Appointment Receipt PDF	EO	12	3	Average	5
Test Request Receipt PDF	EO	10	3	Average	5
Medical Report PDF	EO	15	4	High	7
User Activity Report	EO	12	3	Average	5
Payment Reconciliation Report	EO	14	3	Average	5
Email Notification	EO	6	2	Low	4
SMS Notification	EO	4	2	Low	4
Search Patients	EQ	7	1	Low	3
Search Doctors	EQ	6	2	Low	3
View Appointment Details	EQ	12	3	Average	4
Check Doctor Availability	EQ	4	3	Average	4
View Medical Report	EQ	5	3	Average	4
View Payment History	EQ	4	2	Low	3
View Test Request Status	EQ	4	2	Low	3
View Staff Tasks	EQ	5	2	Low	3
View Department Details	EQ	4	2	Low	3
View User Profile	EQ	2	2	Low	3

Total UFP (Transaction Function)	**155**

Table 6.5 UFP of Data Function

Function Name	Type	DET	RET	Complexity	UFP
Users	ILF	12	1	Low	7
Patients	ILF	13	1	Low	7
Doctors	ILF	12	1	Low	7
Staff	ILF	10	1	Low	7
Departments	ILF	6	1	Low	7
Appointments	ILF	12	1	Low	7
Test Requests	ILF	10	1	Low	7
Reports	ILF	10	1	Low	7
Payments	ILF	10	1	Low	7
Doctor Schedules	ILF	7	1	Low	7
Daily Tasks	ILF	8	1	Low	7
Leaves	ILF	9	1	Low	7
Login Logs	ILF	7	1	Low	7
Payment Gateway (bKash)	EIF	6	1	Low	5
Payment Gateway (Nagad)	EIF	6	1	Low	5
Payment Gateway (Rocket)	EIF	5	1	Low	5
Email Service API	EIF	6	1	Low	5
SMS Gateway API	EIF	5	1	Low	5

Total UFP (Data Function)	**116**

Table 6.6 Total Unadjusted Function Point (UFP)

Function Type	Count	Total UFP
External Input (EI)	16	59
External Output (EO)	11	63
External Inquiry (EQ)	10	33
Internal Logical File (ILF)	13	91
External Interface File (EIF)	5	25
Total Unadjusted Function Points (UFP)	**55**	**271**

6.2.4 Calculation of Total Degree of Influence (TDI)

Table 6.7 Calculation of Total Degree of Influence (TDI)

SI	General System Characteristics	Brief Description	DI
01	Data Communication	System requires online communication between modules, payment gateways (bKash, Nagad, Rocket, DBBL, Mastercard, Visa), email/SMS services, and cloud storage.	5
02	Distributed Data Processing	Data is processed across vendors with cloud hosting, load balancing, and database replication.	3
03	Performance	The system must handle concurrent users efficiently with fast response times (target < 3 seconds), which is critical for hospital operations.	4
04	Heavily Used Configuration	Multiple user roles (patients, doctors, staff, admin) access the system simultaneously with high availability requirements.	4
05	Transaction Rate	High transaction rate due to appointment bookings, payments, staff operations, and concurrent users.	4
06	Online Data Entry	All data entry is performed online through web-based forms with no offline support.	5

07	End-User Efficiency	User-friendly interface with intuitive navigation, mobile responsiveness, and streamlined workflows for medical staff.	5
08	Online Update	Real-time updates for appointments, schedules, payments, inventory, and notifications across all user roles.	5
09	Complex Processing	Includes verification, payment gateway integration, PDF generation, and appointment scheduling algorithms.	4
10	Reusability	Core modules such as authentication, payments, and notifications are reusable across features.	3
11	Installation Ease	The system is designed for easy deployment and configuration on cloud or local servers.	3
12	Operational Ease	Administrative dashboard, automated backups, and monitoring tools simplify daily operations.	4
13	Multiple Sites	Designed for single-hospital deployment with potential for multi-branch expansion.	2
14	Facilitate Change	Modular design supports future upgrades, new features, and vendor expansion.	4

Total TDI	**55**

6.2.5. Final Calculation

Value Adjustment Factor (VAF) = $0.65 + (0.01 \times \text{TDI})$

$= 0.65 + (0.01 \times 55)$

$= 1.10$

Total UFP = UFP (TF) + UFP (DF)

$= 155 + 116$

$= 271$

AFP (Adjusted Function Point) = UFP (Total) * VAF

= 271 * 1.10

= 298.1

6.3 Project Scheduling

We have to complete this project within 12 weeks or 3 Month. So, we have created a Gantt Chart for our use in time management. Our whole timeline is shown in below Gantt Chart.

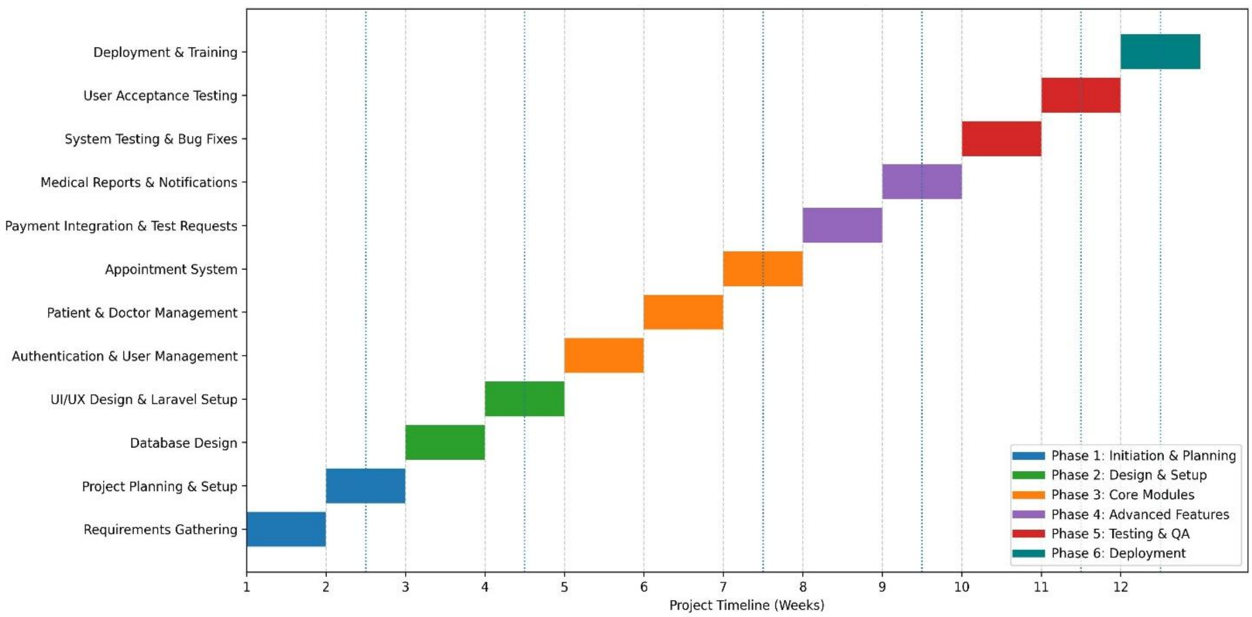


Figure 6.1 Gantt Chart

This Figure 6.1 shows the hospital management system project is divided into six separate phases and has a planned 12-week development timetable. Weeks 1-3 of the project requirement gathering and Laravel setup. Weeks 3-5 are database design UI/UX design and Laravel setup. The appointment system, patient & doctor management are all included in the core development phase (Weeks 5-8). Weeks 8-10 saw the implementation of advance features like payment integration & test request and medical report & notification. System testing and bug fixes, user acceptance testing and deployment and training are the main topics of the last phase (Weeks 10-12). Throughout the project lifecycle are iterative testing and quality assurance are made possible by this staged methodology, which guarantees methodical development with distinct milestone.

Chapter 7.

Designing

7.1 Data Flow Diagram

A Data Flow Diagram (DFD) is a graphical tool used in software engineering to represent how data moves through a system, showing the flow between processes, data stores, and external entities. It helps us to easily identify how system is interchanging its data with database, system and stakeholders.

7.1.1. Level 0 DFD (Context Level)

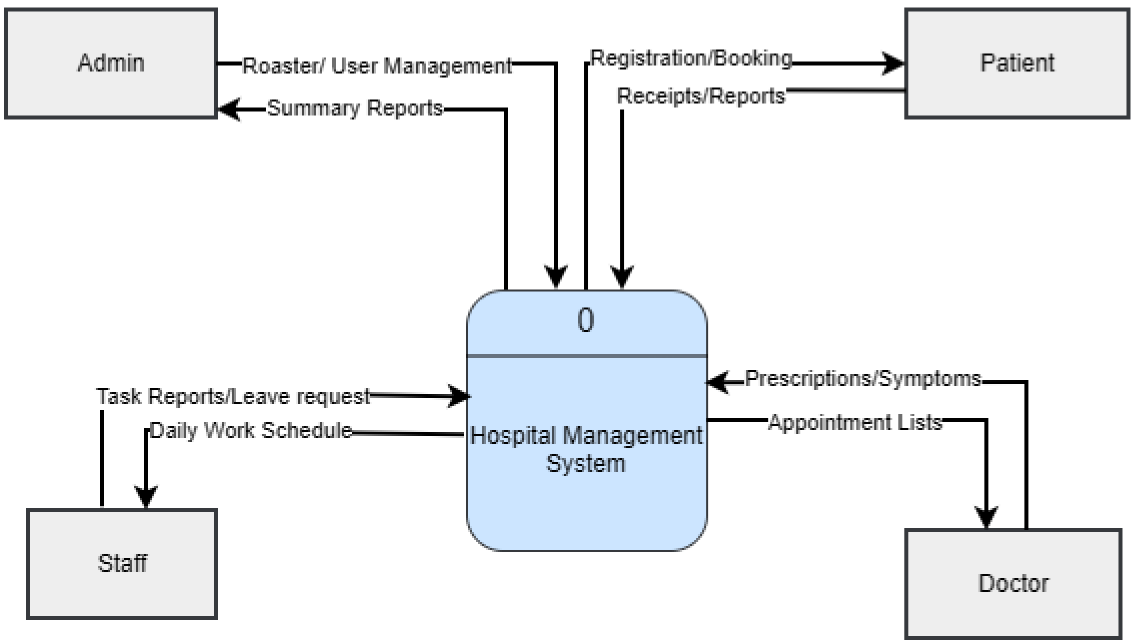


Figure 7.1 DFD Level 0

7.1.2. Level 1 DFD (Process Overview)

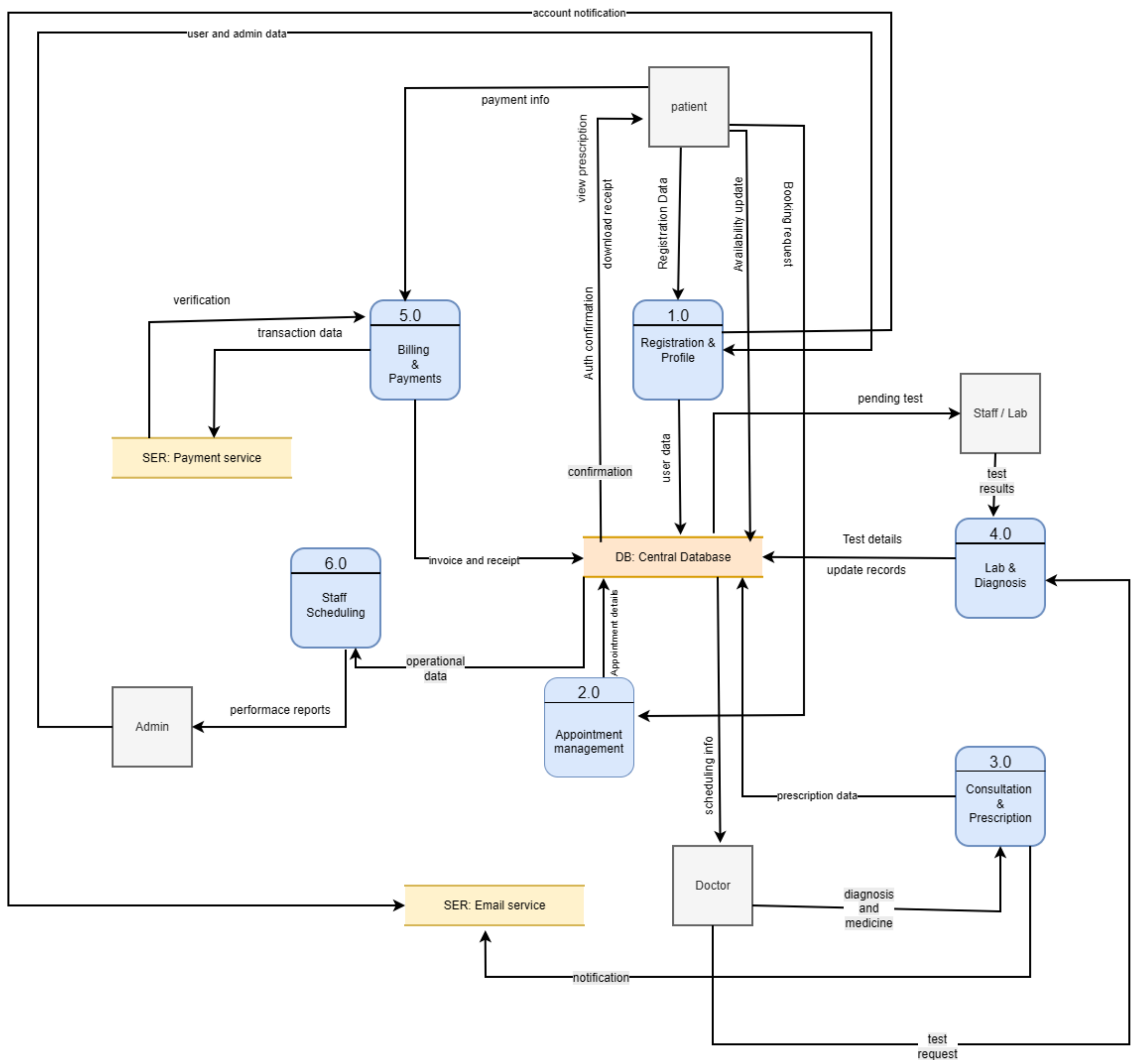


Figure 7.2 DFD Level 1

7.1.3 Level 2 DFD (Registration & Profile Management (Process 1.0))

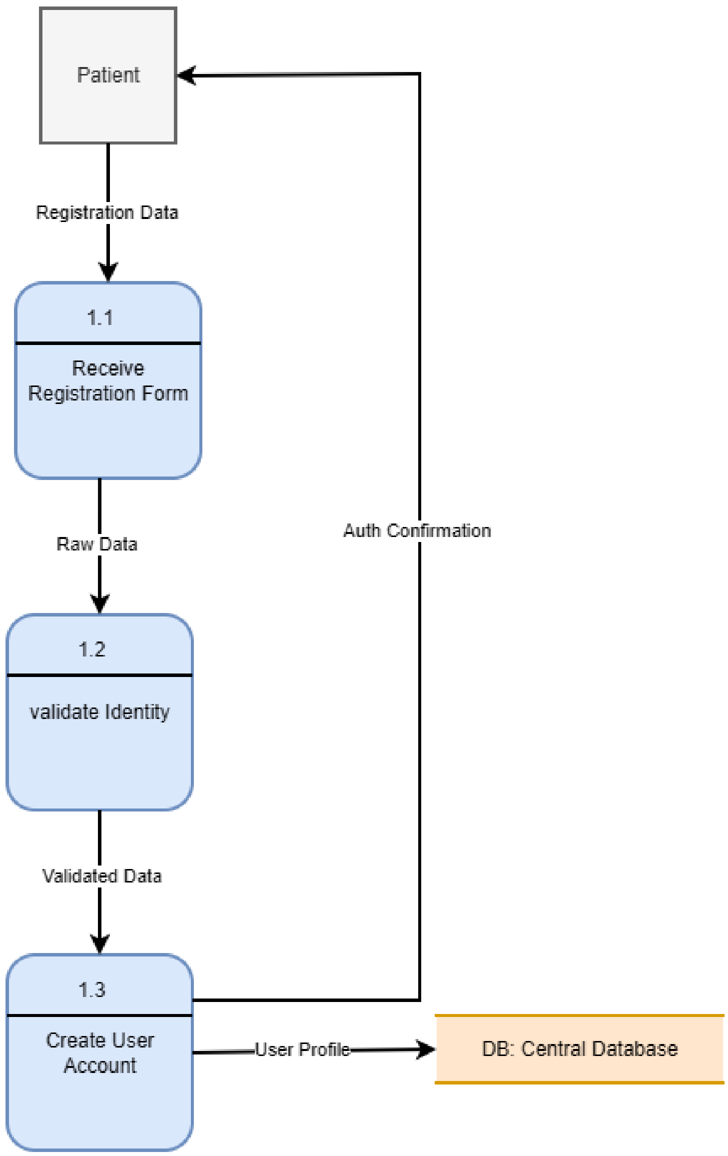


Figure 7.3 Level 2 DFD (Registration & Profile Management (Process 1.0))

7.1.4 Level 2 DFD (Appointment Management (Process 2.0))

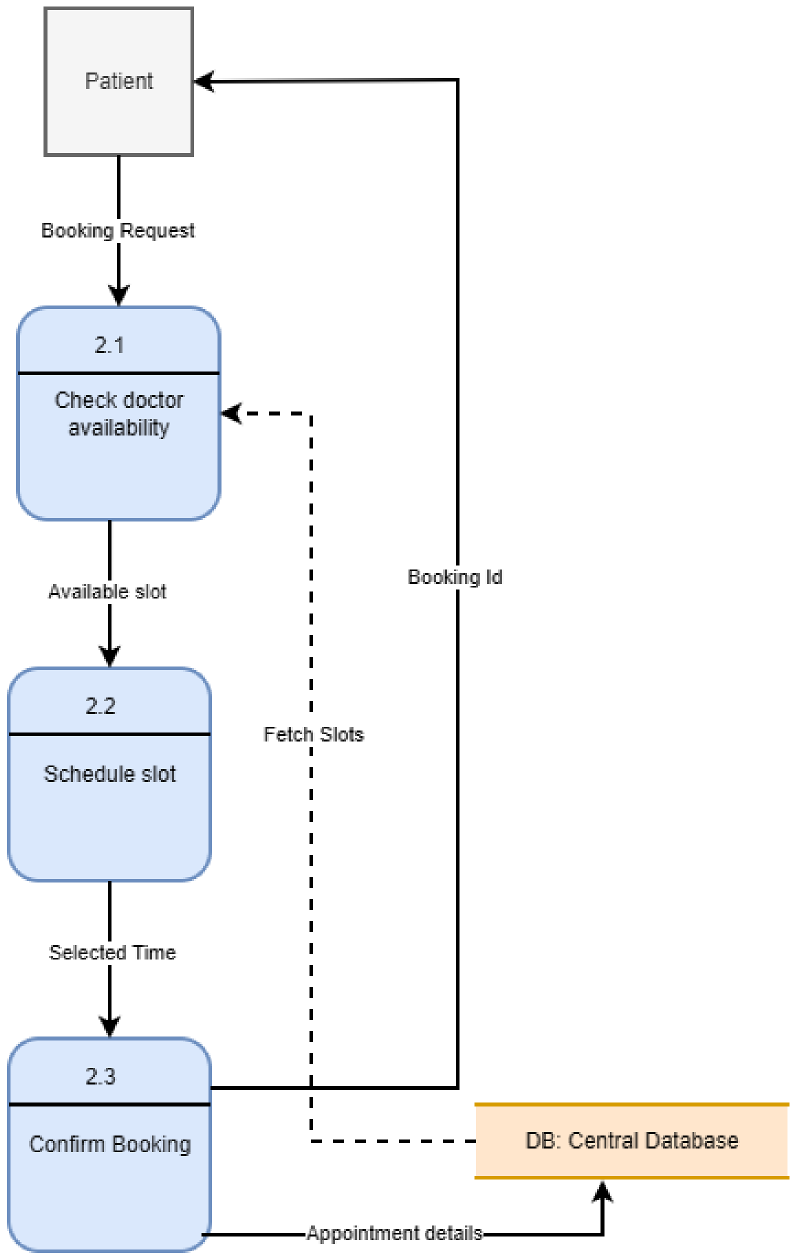


Figure 7.4 Level 2 DFD (Appointment Management (Process 2.0))

7.1.5 Level 1 DFD (Prescription & Consultation (Process 3.0))

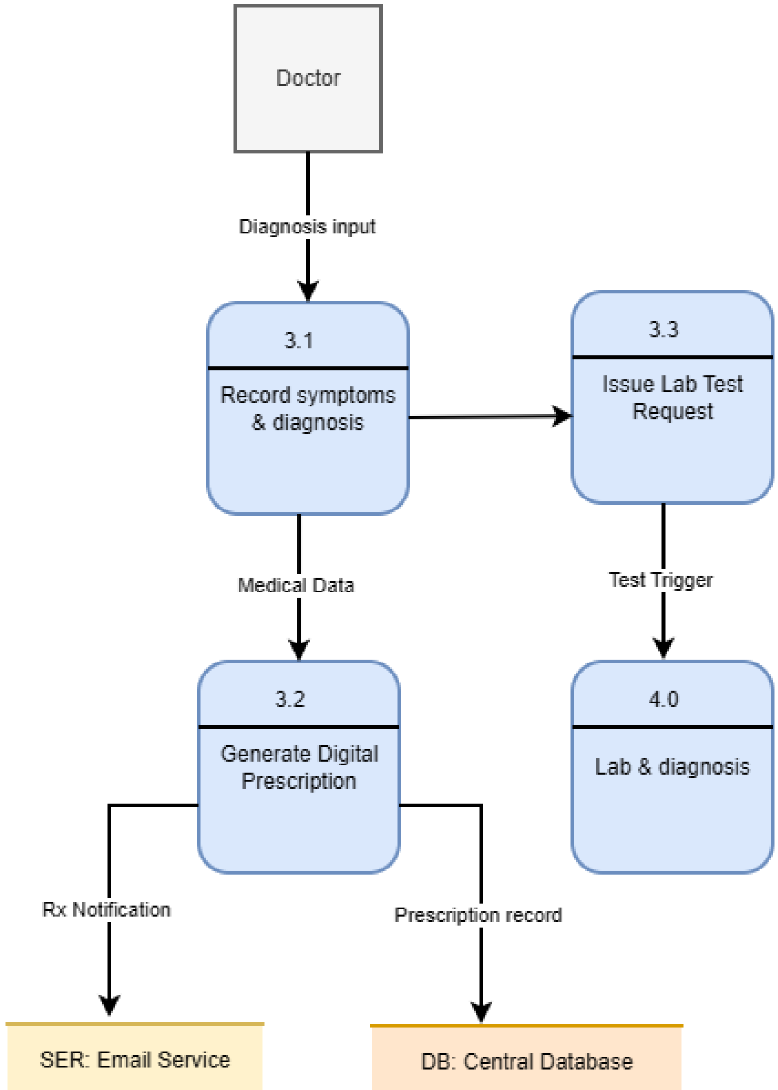


Figure 7.5 Level 1 DFD (Prescription & Consultation (Process 3.0))

7.1.6 Level 1 DFD (Lab & Diagnosis (Process 4.0))

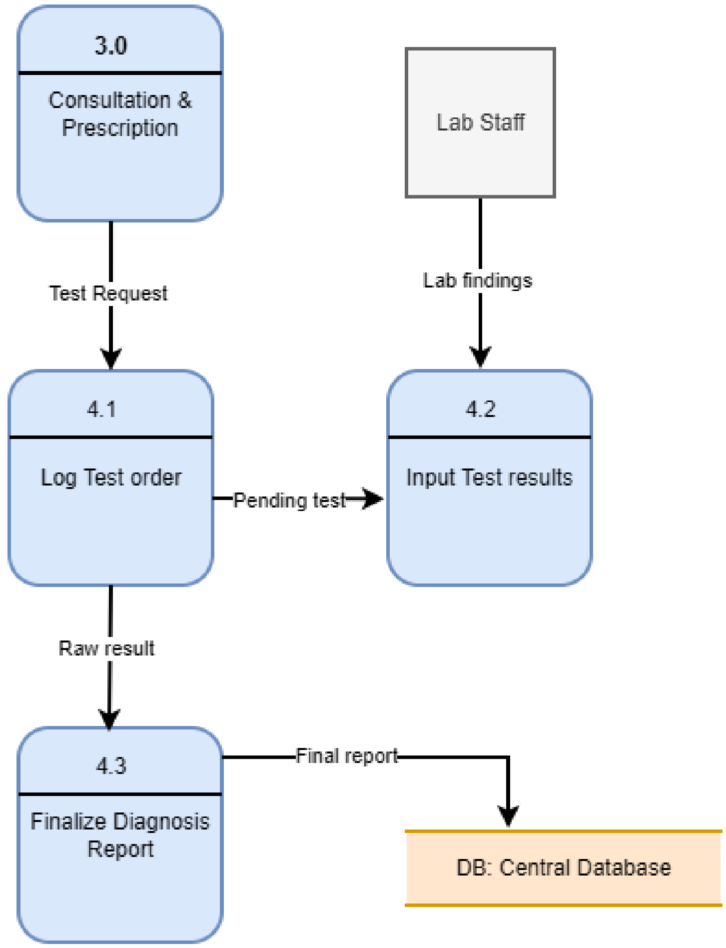


Figure 7.6 Level 1 DFD (Lab Test & Diagnostic Requests (Process 4.0))

7.2 Entity Relationship (ER) Diagram

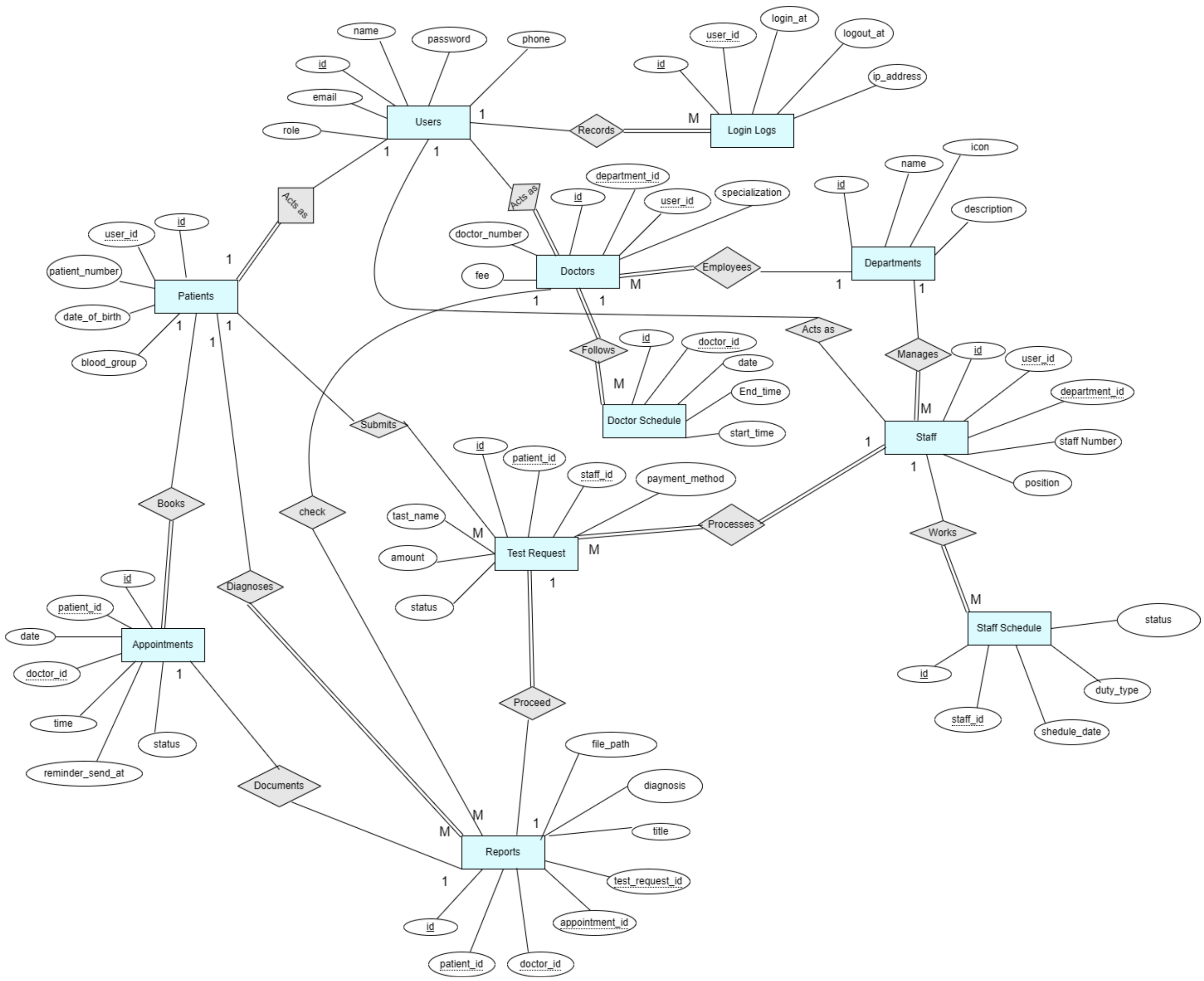


Figure 7.7 Entity Relationship Diagram

7.3 Database Field Design

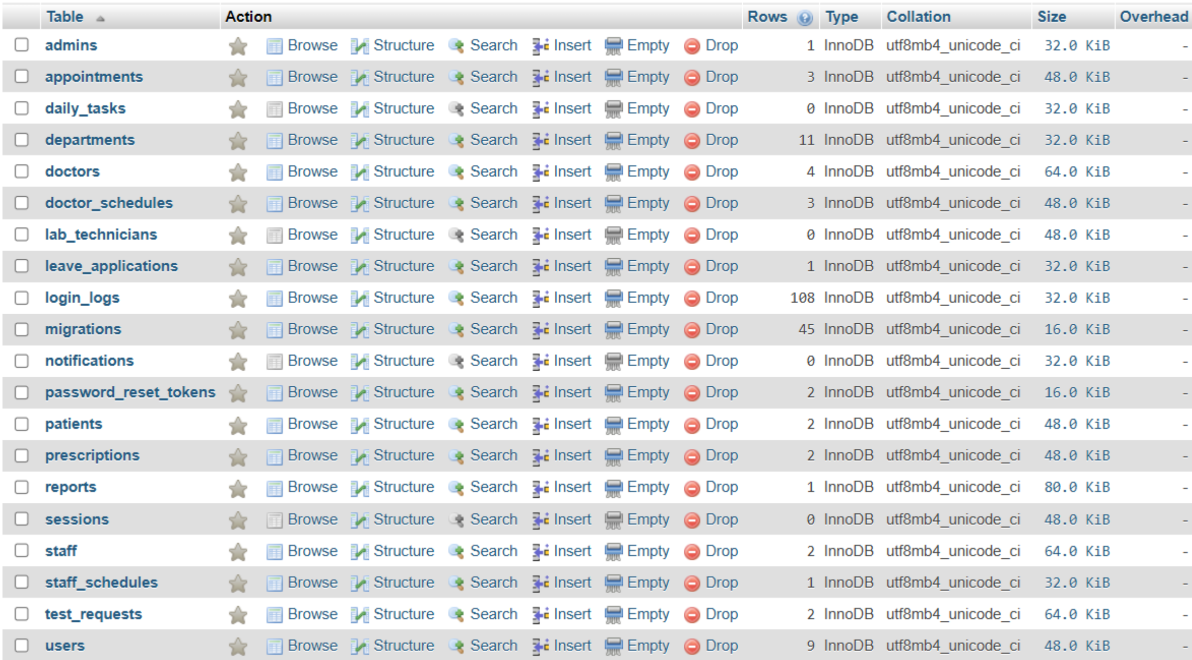


Table	Action	Rows	Type	Collation	Size	Overhead
admins	Browse Structure Search Insert Empty Drop	1	InnoDB	utf8mb4_unicode_ci	32.0 KiB	-
appointments	Browse Structure Search Insert Empty Drop	3	InnoDB	utf8mb4_unicode_ci	48.0 KiB	-
daily_tasks	Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_unicode_ci	32.0 KiB	-
departments	Browse Structure Search Insert Empty Drop	11	InnoDB	utf8mb4_unicode_ci	32.0 KiB	-
doctors	Browse Structure Search Insert Empty Drop	4	InnoDB	utf8mb4_unicode_ci	64.0 KiB	-
doctor_schedules	Browse Structure Search Insert Empty Drop	3	InnoDB	utf8mb4_unicode_ci	48.0 KiB	-
lab_technicians	Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_unicode_ci	48.0 KiB	-
leave_applications	Browse Structure Search Insert Empty Drop	1	InnoDB	utf8mb4_unicode_ci	32.0 KiB	-
login_logs	Browse Structure Search Insert Empty Drop	108	InnoDB	utf8mb4_unicode_ci	32.0 KiB	-
migrations	Browse Structure Search Insert Empty Drop	45	InnoDB	utf8mb4_unicode_ci	16.0 KiB	-
notifications	Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_unicode_ci	32.0 KiB	-
password_reset_tokens	Browse Structure Search Insert Empty Drop	2	InnoDB	utf8mb4_unicode_ci	16.0 KiB	-
patients	Browse Structure Search Insert Empty Drop	2	InnoDB	utf8mb4_unicode_ci	48.0 KiB	-
prescriptions	Browse Structure Search Insert Empty Drop	2	InnoDB	utf8mb4_unicode_ci	48.0 KiB	-
reports	Browse Structure Search Insert Empty Drop	1	InnoDB	utf8mb4_unicode_ci	80.0 KiB	-
sessions	Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_unicode_ci	48.0 KiB	-
staff	Browse Structure Search Insert Empty Drop	2	InnoDB	utf8mb4_unicode_ci	64.0 KiB	-
staff_schedules	Browse Structure Search Insert Empty Drop	1	InnoDB	utf8mb4_unicode_ci	32.0 KiB	-
test_requests	Browse Structure Search Insert Empty Drop	2	InnoDB	utf8mb4_unicode_ci	64.0 KiB	-
users	Browse Structure Search Insert Empty Drop	9	InnoDB	utf8mb4_unicode_ci	48.0 KiB	-

Figure 7.8 Database field design.

7.4 Interface Design

7.4.1 Home page

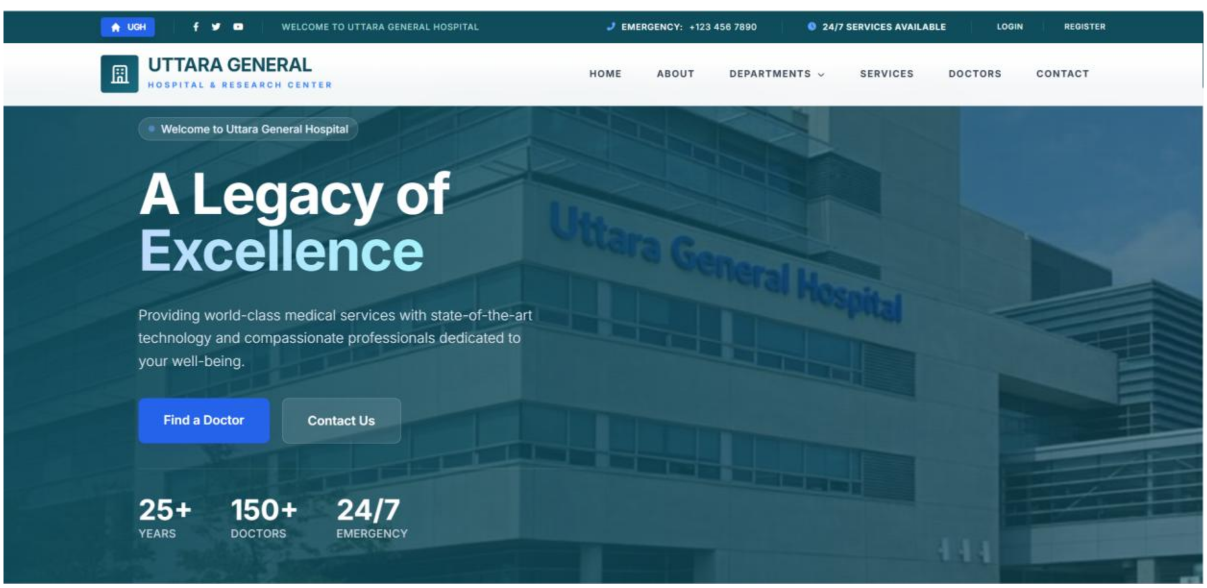


Figure 7.9 Home Page.

7.4.2 Login Page

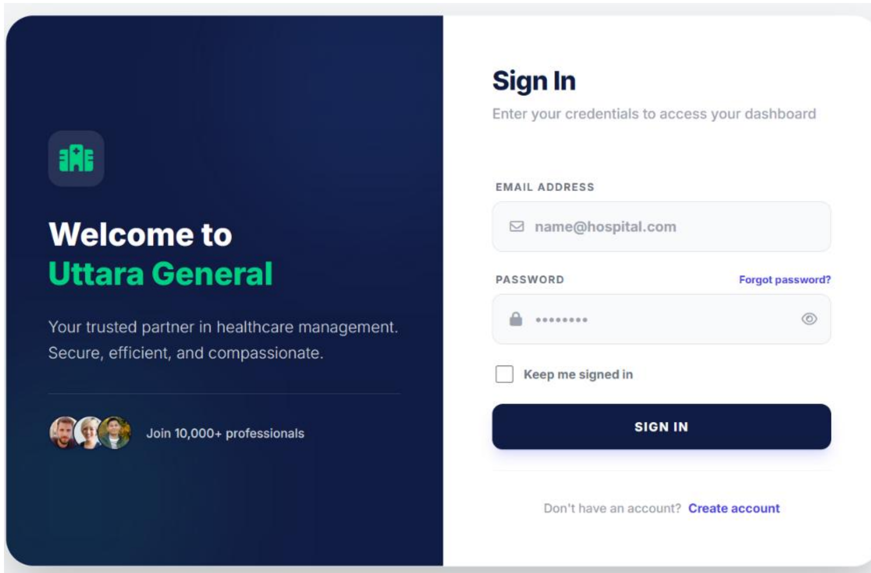


Figure 7.10 User Login Page

7.4.3 Registration Page

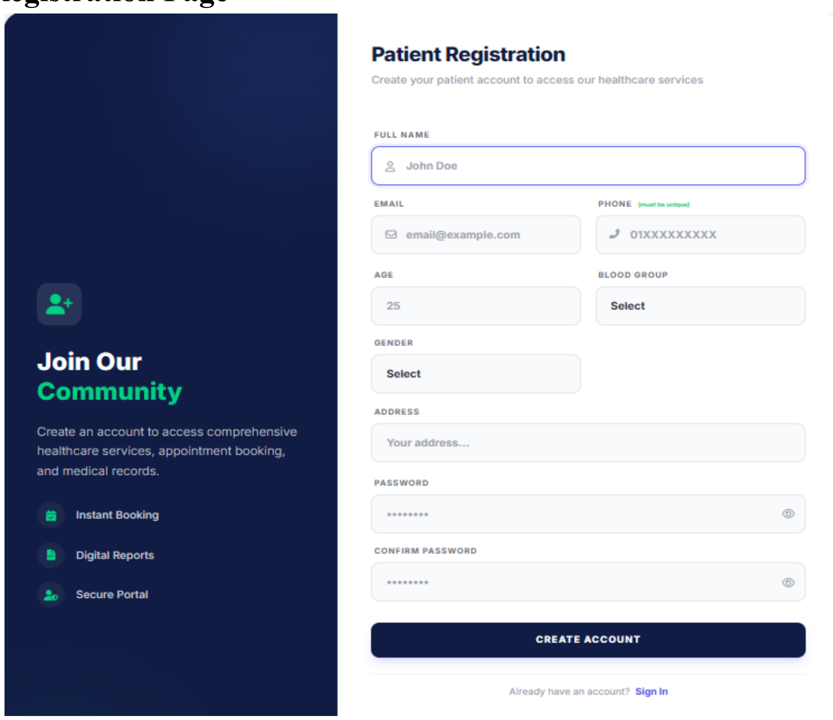


Figure 7.11 Registration Page

7.4.4 Admin Dashboard

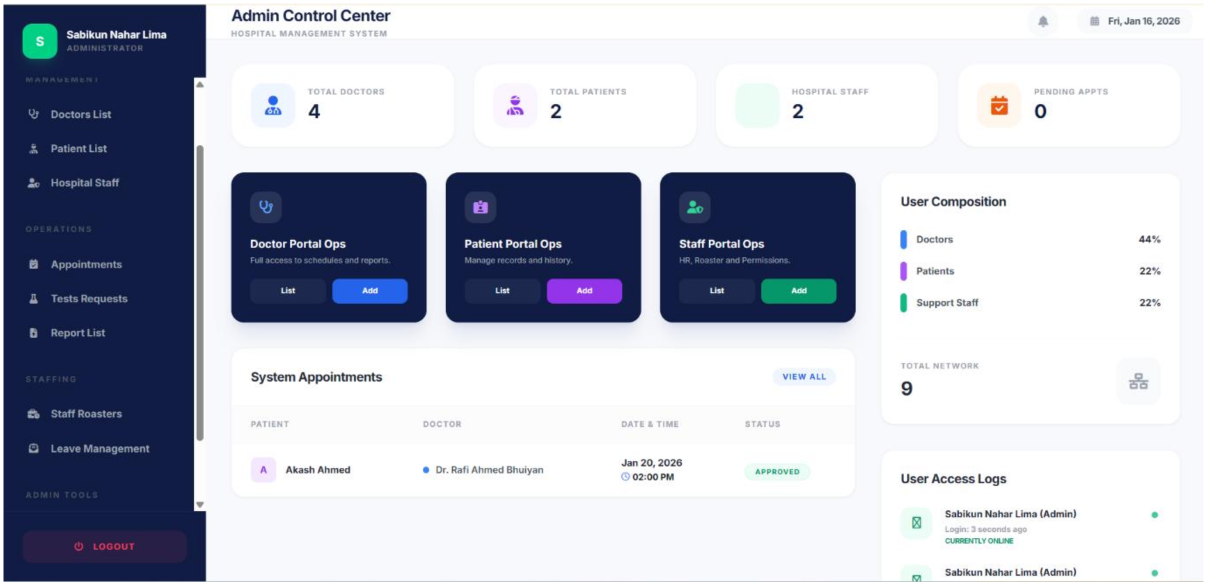


Figure 7.12 Admin Dashboard

7.4.5 Doctor Dashboard

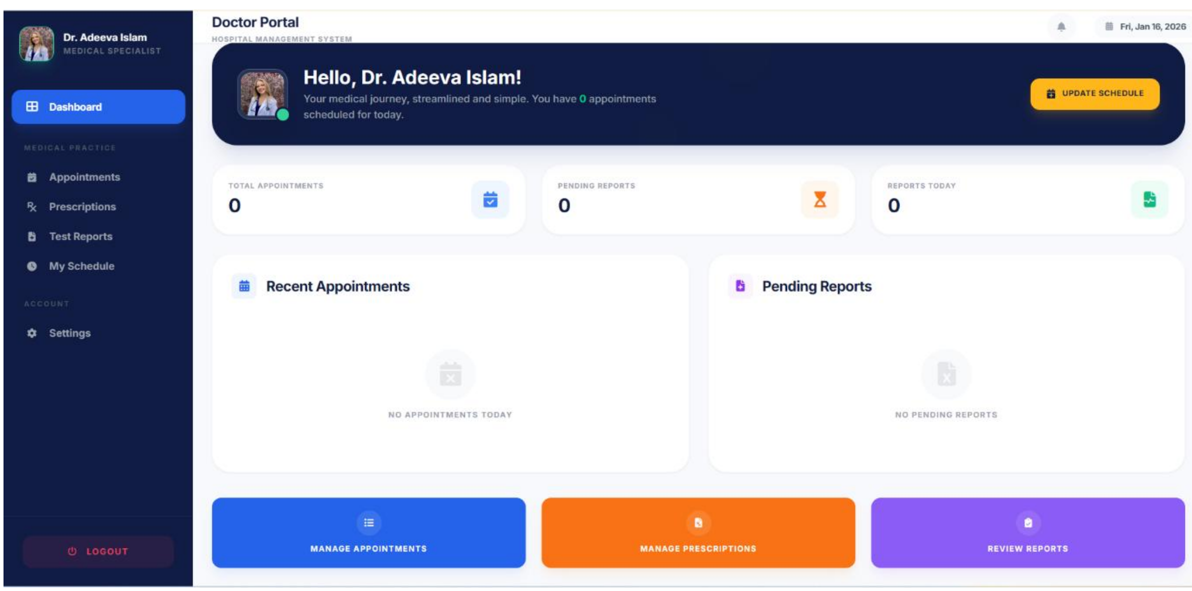


Figure 7.13 Doctor Dashboard

7.4.6 Patient Dashboard

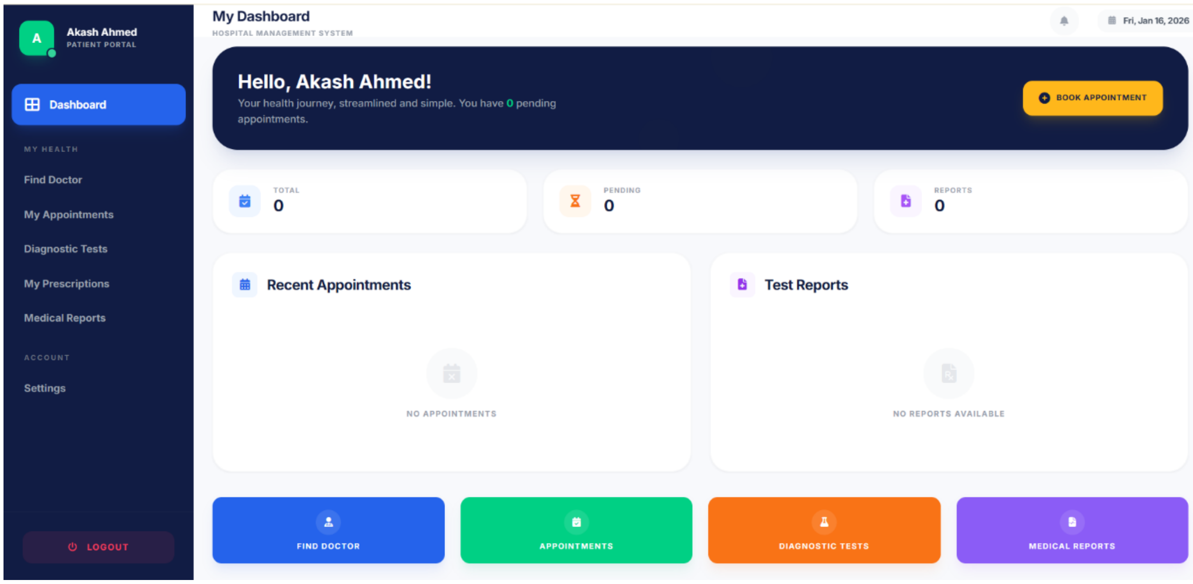


Figure 7.14 Patient Panel

7.4.7 Staff Dashboards

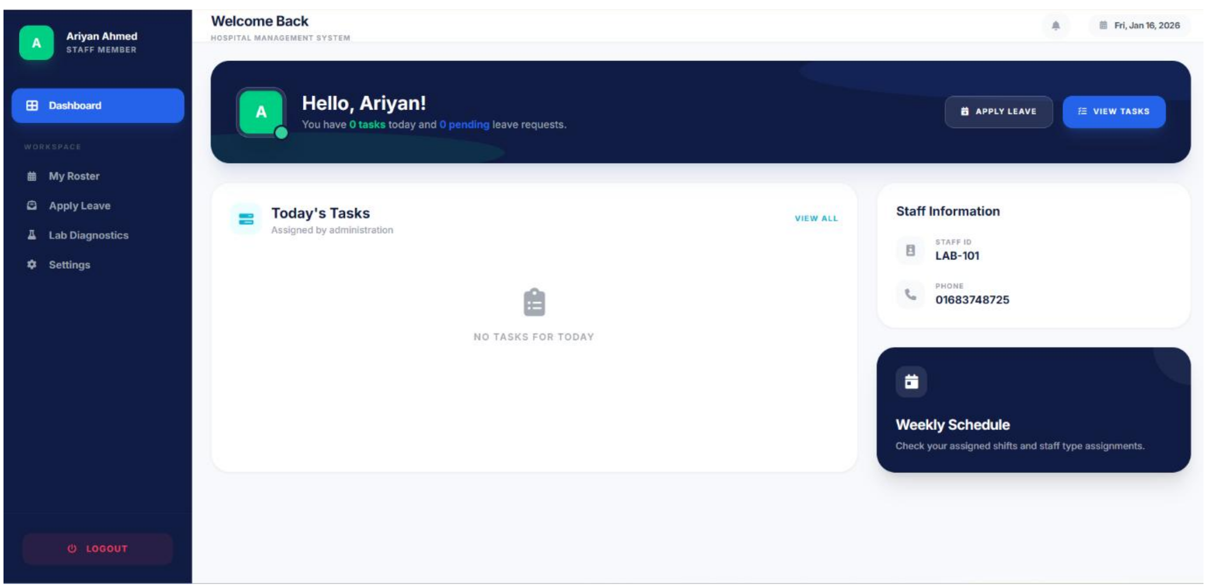


Figure 7.15 Lab Technician Dashboard (Staff)

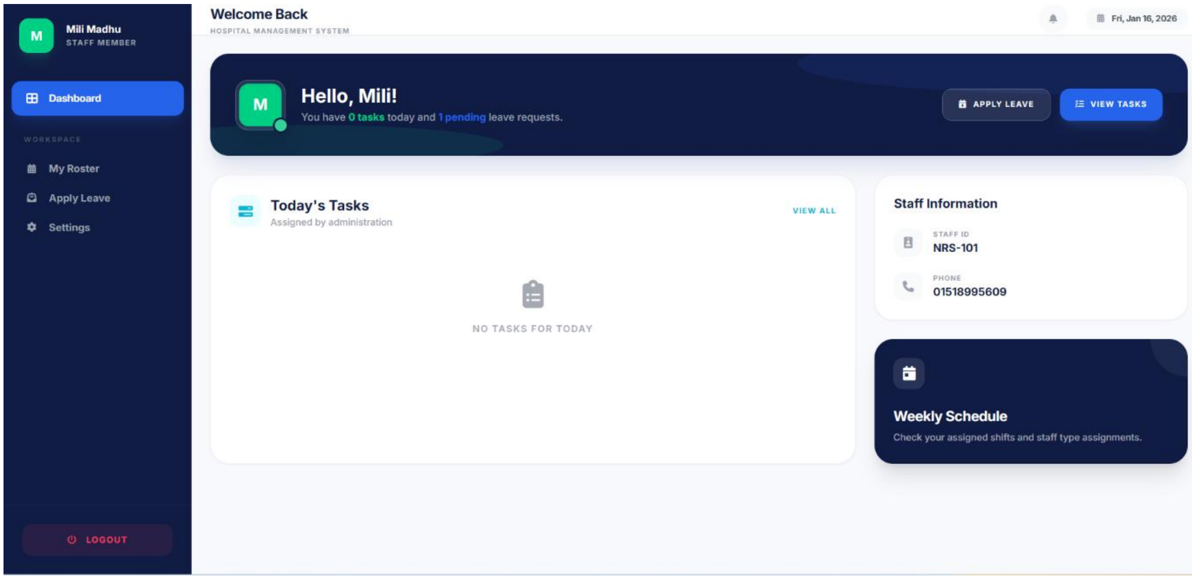


Figure 7.16 Nurse Dashboard (Staff)

7.4.8 Appointment Form

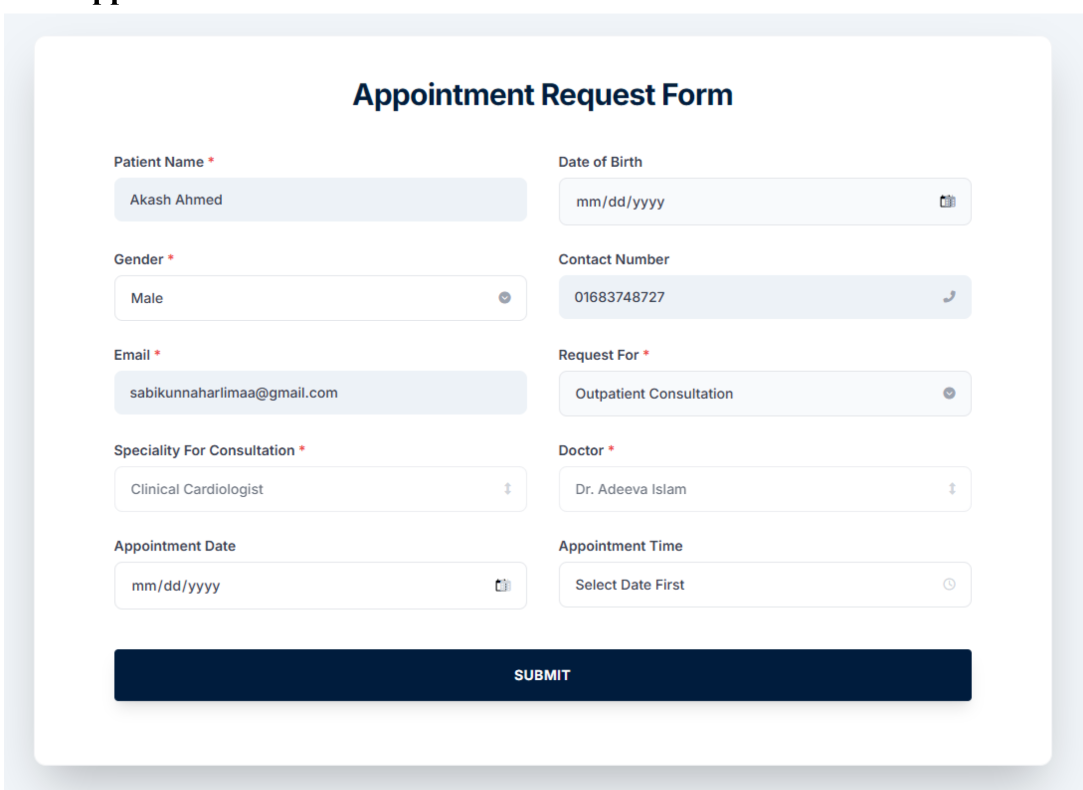


Figure 7.17 Appointment Request Form

7.4.9 Test Request Form

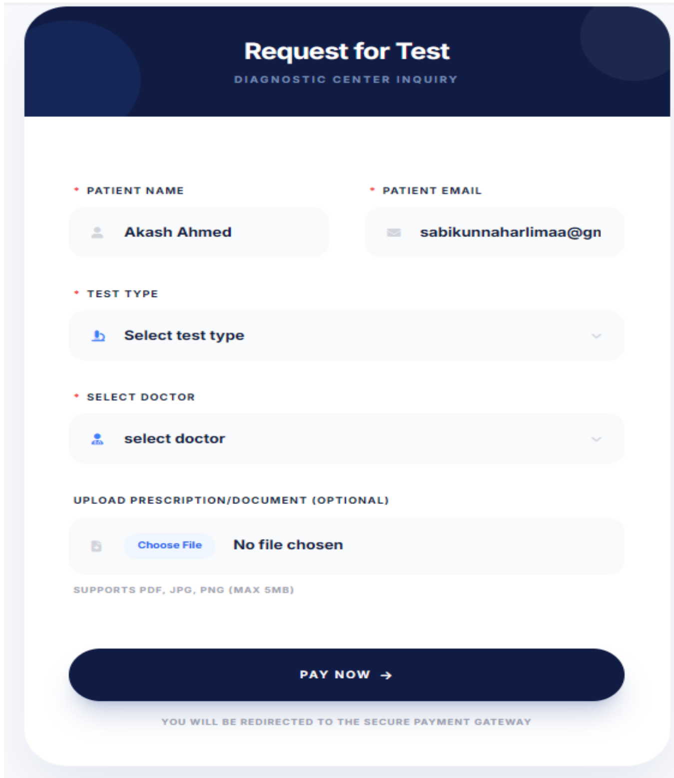


Figure 7.18 Test Request Form

7.4.10 User Activity Report page

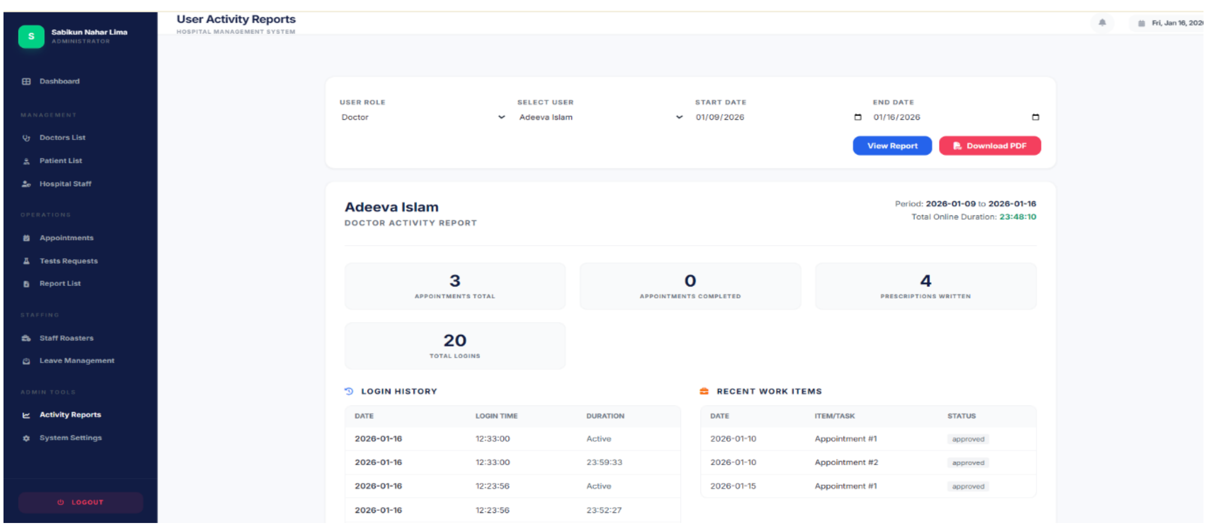
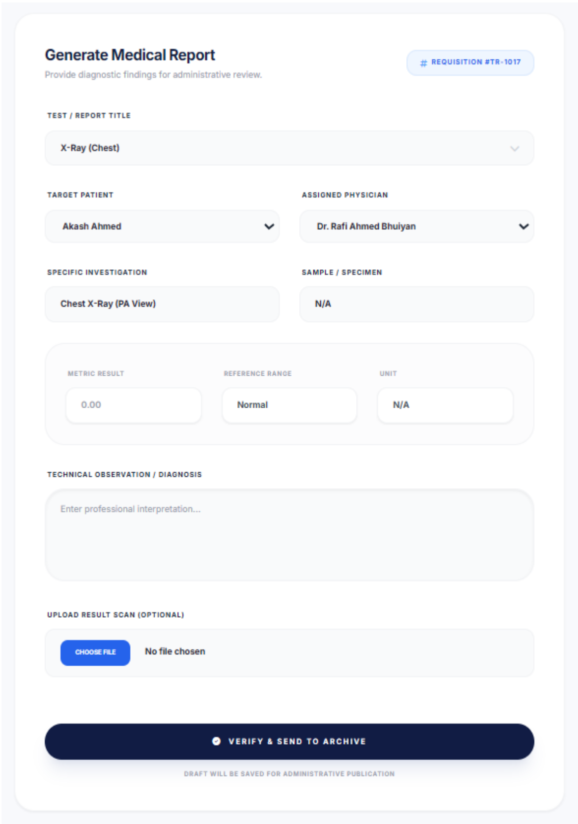


Figure 7.19 User Activity Report Page (Admin Page)

7.4.11 Create Test Report Form and Sample Report



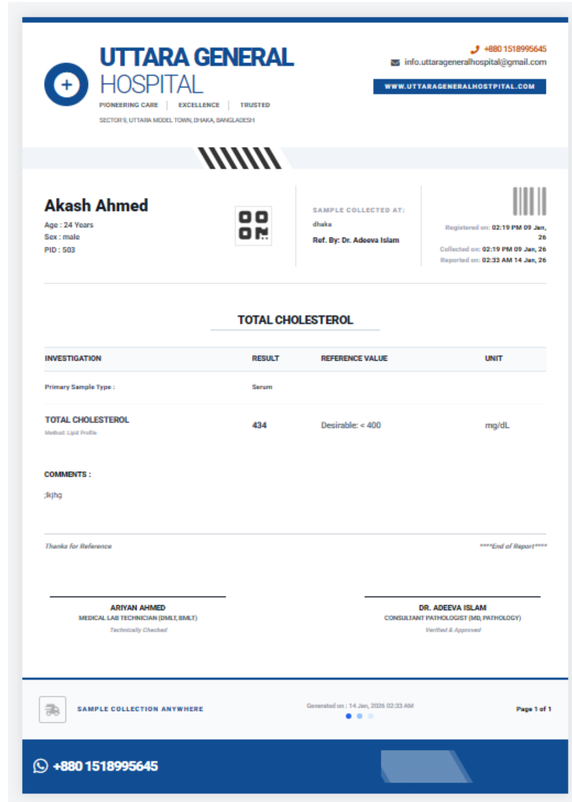


Figure 7.20 Create Test Report Form and Sample Report

7.4.12 Prescription Form

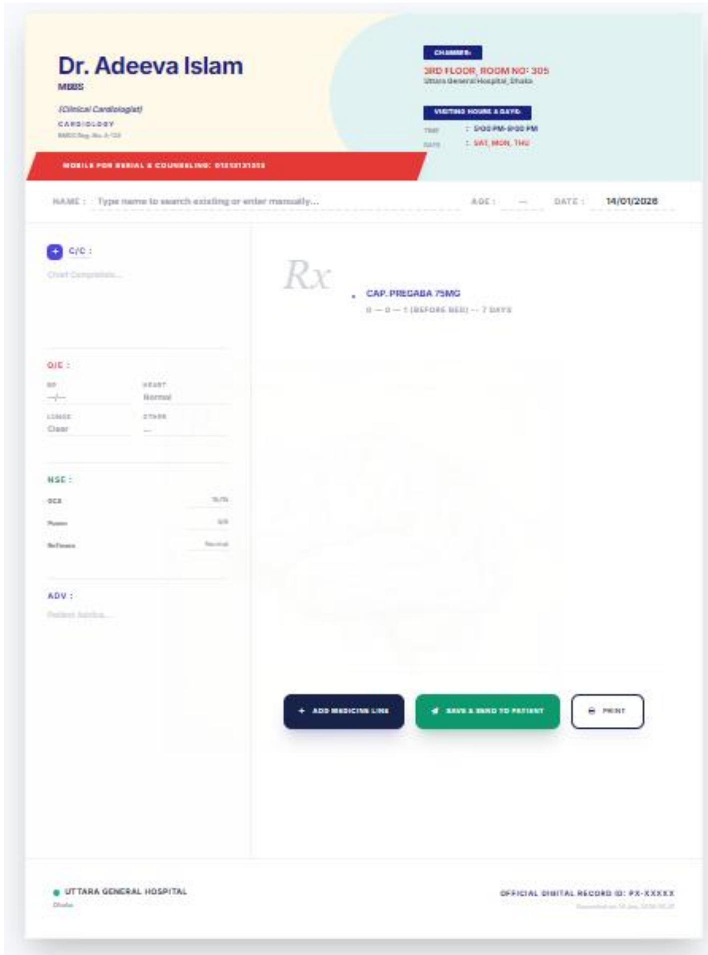


Figure 7.21 Prescription Form

Chapter 8.

Testing

8.1 Test Cases

A set of conditions or inputs designed to check whether a software feature or function works correctly or not is called test case. A test case includes test data, execution steps, and the expected outcome. These test cases are used to verify that the system meets its requirements, identify any bug, and it ensure that the software is reliable and functions correctly.

Table 8.1 Test Case 1.

<table>
<tr><td colspan="3">Test Case ID: 1
Test Priority: High
Module Name: Patient Registration
Test Title: New Patient Registration Test</td><td colspan="4">Test Designed By: Sabikun Nahar Lima
Test Designed Date: 12-01-2026
Test Executed By: Sabikun Nahar Lima</td></tr>
<tr><td colspan="7">Description: Prospective patient can successfully create an account.</td></tr>
<tr><td colspan="7">Pre-Condition: Prospective patient can successfully create an account.</td></tr>
<tr><td colspan="7">Dependencies: Database Connection</td></tr>
<tr><td>Steps</td><td>Test Steps</td><td>Test Data</td><td>Expected Result</td><td>Actual Result</td><td>Status</td><td>Note</td></tr>
<tr><td>1</td><td>Click "Register" button</td><td rowspan="3">Name:
Akash Ahmed Email:
sabikunnaharlimaa@gmail.com
Pass: Akash@123</td><td rowspan="3">Successfully Registered and user will be redirected to dashboard</td><td rowspan="3">Successfully Registered and user is redirected to dashboard</td><td rowspan="3">Pass</td><td rowspan="3"></td></tr>
<tr><td>2</td><td>Enter Registration Data</td></tr>
<tr><td>3</td><td>Click "Register"</td></tr>
</table>

Table 8.2 Test Case 2

<table>
<tr><td colspan="4">Test Case ID: 2
Test Priority: High
Module Name: User Login
Test Title: Role-Based Login Access Test</td><td colspan="3">Test Designed By: Sabikun Nahar Lima
Test Designed Date: 12-01-2026
Test Executed By: Sabikun Nahar Lima</td></tr>
<tr><td colspan="7">Description: Different users (Admin, Doctor, Patient) can access their respective panels.</td></tr>
<tr><td colspan="7">Pre-Condition: Registered accounts in the system.</td></tr>
<tr><td colspan="7">Dependencies: Authentication Middleware</td></tr>
<tr><td>Steps</td><td>Test Steps</td><td>Test Data</td><td>Expected Result</td><td>Actual Result</td><td>Status</td><td>Note</td></tr>
<tr><td>1</td><td>Enter Admin Email/Pass</td><td rowspan="3">Email: nahar.sabikunlima@gmail.com
Password: password0987</td><td rowspan="3">Access Patient Dashboard</td><td rowspan="3">Patient Dashboard</td><td rowspan="3">Pass</td><td rowspan="3"></td></tr>
<tr><td>2</td><td>Enter Patient Email/Pass</td></tr>
<tr><td>3</td><td>Click Logout</td></tr>
</table>

Table 8.3 Test Case 3

Test Case ID: 3 Test Priority: High Module Name: Appointment Booking Test Title: Patient Appointment Request Test			Test Designed By: Sabikun Nahar Lima Test Designed Date: 13-01-2026 Test Executed By: Sabikun Nahar Lima			
Description: Patient can request an appointment with a specific doctor.						
Pre-Condition: Patient logged in.						
Dependencies: Doctor Schedules Table						
Steps	Test Steps	Test Data	Expected Result	Actual Result	Status	Note
1	Select Doctor & Dept	Dept: Cardiology, Doctor: Dr. Adeeva Islam	Slot shown based on schedule	Slots displayed	Pass	
2	Select Date & Time					
3	Submit Request					

Table 8.4 Test Case 4

<table>
<tr><td colspan="3">Test Case ID: 4

Test Priority: Medium

Module Name: Billing & Invoicing

Test Title: Staff Generation of Test Receipt Test</td><td colspan="4">Test Designed By: Sabikun Nahar Lima

Test Designed Date: 13-01-2026

Test Executed By: Sabikun Nahar Lima</td></tr>
<tr><td colspan="7">Description: Staff can generate a bill/receipt for medical tests.</td></tr>
<tr><td colspan="7">Pre-Condition: Staff logged in, pending test request exists.</td></tr>
<tr><td colspan="7">Dependencies: Test Requests Table</td></tr>
<tr><td>Steps</td><td>Test Steps</td><td>Test Data</td><td>Expected Result</td><td>Actual Result</td><td>Status</td><td>Note</td></tr>
<tr><td>1</td><td>Locate Test Request</td><td rowspan="3">Patient ID: 005

Amount: 500.00, Method: bkash</td><td rowspan="3">System records payment</td><td rowspan="3">Recorded</td><td rowspan="3">Pass</td><td rowspan="3"></td></tr>
<tr><td>2</td><td>Enter Payment Info</td></tr>
<tr><td>3</td><td>Click "Generate PDF"</td></tr>
</table>

Table 8.5 Test Case 5

<table>
<tr><td colspan="3">Test Case ID: 5
Test Priority: High
Module Name: Medical Reports
Test Title: Admin Report Upload & Patient Download Test</td><td colspan="4">Test Designed By: Sabikun Nahar Lima
Test Designed Date: 14-01-2026
Test Executed By: Sabikun Nahar Lima</td></tr>
<tr><td colspan="7">Description: Admin can upload a report and Patient can download it.</td></tr>
<tr><td colspan="7">Pre-Condition: Signed medical report file ready.</td></tr>
<tr><td colspan="7">Dependencies: File Storage (Public path)</td></tr>
<tr><td>Steps</td><td>Test Steps</td><td>Test Data</td><td>Expected Result</td><td>Actual Result</td><td>Status</td><td>Note</td></tr>
<tr><td>1</td><td>Upload Report (Admin)</td><td rowspan="3">File: health_report.pdf
Login as Patient</td><td rowspan="3">File saved to server</td><td rowspan="3">Success</td><td rowspan="3">Pass</td><td rowspan="3"></td></tr>
<tr><td>2</td><td>View Reports (Patient)</td></tr>
<tr><td>3</td><td>Click "Download"</td></tr>
</table>

Chapter 9.

Ethical Consideration and Sustainability

9.1 Ethical Considerations

Fundamentals ethical values served as the foundation for the creation of this hospital management system. Healthcare is the main focus. This system is built to benefit healthcare providers and patients alike without creating damage. Transparency in how appointment slots are allocated and the integrity of medical data are the pillars of the system's ethical framework.

9.1.1 Data Privacy and Security

Strong security the system manages protected health information (PHI):

a. Authentication: Bycrpt password hashing is used with Laravel safe authentication.

b. Authorization: By putting role-based access control (RBAC) into place, it is ensured that a patient cannot access admin settings and a staff member cannot examine a doctor’s clinical notes.

c. Integrity: Database are used to guard against data corruption when scheduling.

d. Confidentiality: To maintain privacy.

9.1.2. Intellectual Property

By using open-source framework (Laravel, PHP) according to their license, the project protects intellectual property. The developer keeps ownership of all established code, including the particular logic for automated billing and doctor scheduling. This implementation has not made use of any unapproved third-party libraries.

9.1.3. User Impact

In the development of this system, strong emphasis was placed on user impact to ensure proper and effective usage. As a result, a user-friendly interface was designed to manage all activities within a single platform. Each user is provided with an individual dashboard enriched with well-categorized features. The system mainly benefits two user groups patients and healthcare providers by enabling 24/7 healthcare services in a paperless environment.

9.1.4. Bias and fairness

During the development of the system, a core principle was ensuring fairness and equality for all users. The system was designed to provide services without discrimination or unfair practices. All users can access and use the system without barriers such as language, personal preferences, or other limitations. If any algorithmic bias is identified, it will be promptly addressed and corrected. The ultimate goal of this system is to ensure equal access and opportunities for every user.

9.1.5. Responsibility to Stakeholders

The main stakeholders of the system are Admin, Doctors, Staff, and Patients. Throughout the system development, the highest priority was given to meeting stakeholder needs. All requirements suggested by stakeholders were carefully considered and implemented. Continuous feedback was collected from stakeholders to improve the system. This approach helps build trust and ensures the development of a product that supports long-term success for everyone involved. The system maintains a sense of accountability to its stakeholders:

a. To Hospital Admin: Providing accurate financial and activity reports for better decision-making.

b. To Doctors: Respecting their schedules by preventing over-booking.

c. To Patients: Ensuring their digital records are accurate and always accessible.

d. To Staff: Ensuring their Roaster are timely provided and they can apply for leave.

9.1.6. Professional Responsibility

While developing the system, I maintained professional responsibility by working honestly, carefully, and with a continuous improvement mindset. Best practices were followed to ensure the system is user-friendly and reliable. A well-structured design was maintained so future updates can be made without disruption. Clear documentation and records were kept to support easy understanding and maintenance by other developers. Proper security measures were applied to keep the system safe and robust. Any issues were communicated responsibly and resolved promptly. Additionally, all third-party and copyrighted libraries were properly cited.

9.2 Sustainability Through Software Engineering

The projects sustainability is ensured by code optimization. Server-side resource usage is decreased by employing strong relationships in database queries. The technology eliminates the need for redundant hardware and high-energy cooling systems, that are normally needed for large, unorganized physical server rooms.

9.2.1. Economical Sustainability

Economic sustainability refers to developing a system within the project budget while minimizing overall costs. Achieving this requires proper planning, risk management, accurate cost estimation, and a well-defined timeline, all of which were maintained throughout this project. Economic sustainability also considers whether the system is cost-effective for users. As the system follows a multi-vendor SaaS model, it allows vendors to purchase and use the platform at a minimal cost. Additionally, the system was designed to be user-friendly and built on standardized practices that do not require frequent changes. These factors ensure the economic sustainability of the system and support its long-term growth.

9.2.2. Social Sustainability

Social sustainability focuses on how software impacts individuals, communities, and society over the long term. It goes beyond writing code and emphasizes building a product that promotes trust and supports social well-being. During the development of this system, strong consideration was given to protecting user privacy, security, and individual rights. Fair work practices were also encouraged to ensure developers and teams operate in healthy and balanced environments without excessive workload. Ultimately, the system aims to address real-world problems and contribute positively to society without causing negative social impacts.

Chapter 11.

Conclusion

10.1 Conclusion

The manual, paper-based management is very much problematic there was chances of lack of patient record, manage schedule etc. The system offers a reliable, safe and expendable solution for handling complicated healthcare operations by using the MySQL database and Laravel framework. The project provides a smooth information flow by implementing separate modules for Admins, Doctors, Staff and Patients. Eventually, the system not only improves the operational efficiency of the hospital but also greatly improves the patient experience.

10.2 Limitations

There are still some limitations that are not part of the projects original scope, considering many features that were put in place:

a. No real-time communication: Because this system not have any instant messaging and live chat option.

b. Wen only access: this system is just web based not have any mobile application for iOS or Android.

c. Hardware Integration: in this system just online based that re not have any physical equipment.

d. Limited telehealth: this system is not fully online based; I have imagination of physical hospital that's why I am using just online platform so there is no video conferencing.

e. Language Support: This System only have English language so this can be problematic for some localized environment.

10.3 Future Plans

In the future the system will include the following improvements to better develop it:

a. Using AI: Using Machine Learning algorithms is to assist doctors by examine historical patient data to predict potential health risks.

b. Pharmacy & Inventory: Adding a new module pharmacy management that tracks medicine stock and automatically updates inventory when a doctor writes a prescription.

c. Mobile Application: Developing a cross-platform mobile app (using Flutter or React Native) to provide patients easy services. Patient can easily take services from the system.

d. Multi-Language Support: Adding a localization multi-language feature to allow users to use different languages (Bengali and English) for better understanding.

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